Output Impedance (Z_o)

To find the output impedance we have to set V_i to zero. Note that $V_i = V_{gs}$. Hence, with $V_i = 0$, $g_m V_{gs} = 0$. Thus the current source $g_m V_{gs}$ is represented by an open circuit as shown in Fig. 9.11.

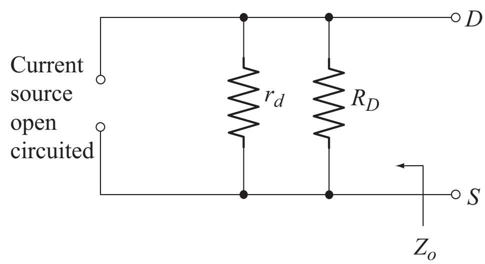


Fig. 9.11 Circuit to find Z_o

From the circuit of Fig. 9.11, the output impedance Z_o is given by

$$Z_o = r_d \,||\, R_D \tag{9.19}$$

Voltage Gain (A_V)

$$A_V = \frac{V_o}{V_i} \tag{9.20}$$

From Fig. 9.10,

$$V_i = V_{gs} \tag{9.21}$$

The output circuit of Fig. 9.10 is redrawn in Fig. 9.12 for convenience.

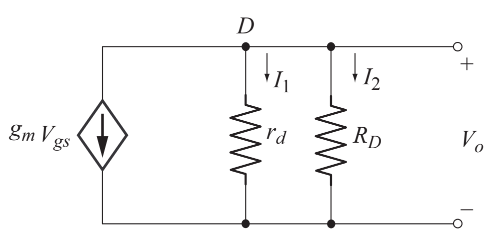


Fig. 9.12 Output ac equivalent Circuit

Appling KCL at the node D we have

$$g_m V_{gs} + I_1 + I_2 = 0$$

$$I_1 = \frac{V_o}{r_d} \quad \text{and} \quad I_2 = \frac{V_o}{R_D}$$

Using these relations in the above equation, we get

$$g_m V_{gs} + \frac{V_o}{r_d} + \frac{V_o}{R_D} = 0$$

$$g_m V_{gs} = -V_o\left[\frac{1}{r_d} + \frac{1}{R_D}\right]$$

$$g_m V_{gs} = -V_o\left[\frac{r_d + R_D}{r_d R_D}\right]$$

$$V_o = -g_m V_{gs}\left[\frac{r_d R_D}{r_d + R_D}\right]$$

or $$V_o = -g_m V_{gs}\,[r_d \,||\, R_D] \tag{9.22}$$

Using Equations (9.21) and (9.22) in Equation (9.20), we have

$$A_V = \frac{-g_m V_{gs}\left[r_d \,||\, R_D\right]}{V_{gs}}$$

$$A_V = -g_m\,[r_d \,||\, R_D] \tag{9.23}$$

The negative sign in Equation (9.23) reveals that, V_i and V_o are 180° out of phase. Note that common-source configuration is an inverting amplifier.

For $r_d \geq 10\,R_D$

When $r_d = 10\,R_D$

$$r_d \,||\, R_D = 0.909\,R_D$$

Hence when $r_d \geq 10\,R_D$, we can take

$$r_d \,||\, R_D \approx R_D$$

From Equation (9.19)

$$Z_o \approx R_D \tag{9.24}$$

From eqution (9.23)

$$A_V \approx -g_m R_D \tag{9.25}$$

Important characteristics of common-source configuration

Based on the results derived, we can attribute the following characteristics to common-source configuration

- High input impedance [$Z_i \approx R_G$]
- Medium high output impedance [$Z_o \approx R_D$]
- Medium voltage gain [$A_V \approx -g_m R_D$]
- 180° phase difference between input and output voltages.

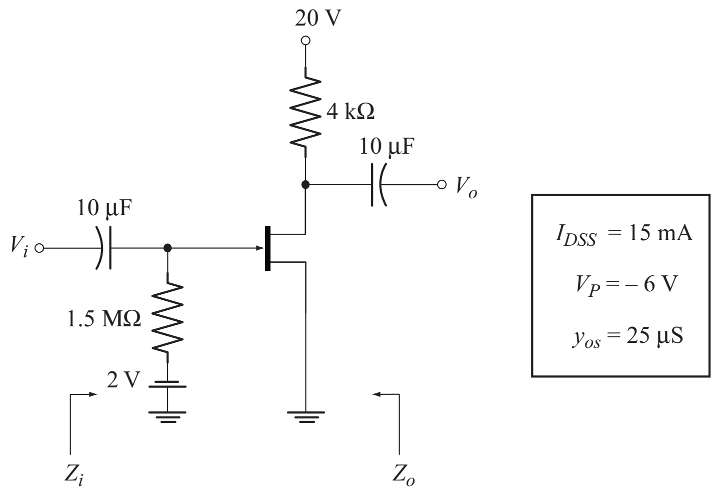
Example 9.3

For the FET amplifer shown below:

(a) Calculate Z_i and Z_o

(b) Calculate A_V

(c) Calculate Z_i, Z_o and A_V, neglecting the effect of r_d and compare the results.



Solution

First we have to find g_m and r_d

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{2(15\text{ mA})}{6\text{V}} = 5\text{ mS.}$$

$$V_{GS_Q} = -V_{GG} = -2\text{ V}$$

$$g_m = 5\text{ mS}\left[1-\frac{-2\text{V}}{-6\text{V}}\right] = 3.33\text{ mS}$$

$$r_d = \frac{1}{y_{o_s}} = \frac{1}{25\,\mu\text{S}} = 40\text{ k}\Omega.$$

(a) $$Z_i = R_G = 1.5\text{ M}\Omega$$

$$Z_o = r_d \,||\, R_D = 40\text{ k}\Omega \,||\, 4\text{ k}\Omega = 3.63\text{ k}\Omega.$$

(b) $$A_V = -g_m[r_d \,||\, R_D] = -[3.33\text{ mS}][3.63\text{ k}\Omega] = -12.08$$

(c) $$r_d = 40\text{ k}\Omega \quad\text{and}\quad 10\,R_D = 40\text{ k}\Omega.$$

Since $r_d = 10\,R_D$, we can ignore the effect of r_d. Neglecting r_d we have:

$$Z_i = R_G = 1.5\text{ M}\Omega$$

$$Z_o \approx R_D = 4\ \text{k}\Omega$$
$$A_V \approx -g_m R_D = -[3.33\ \text{mS}]\ [4\ \text{k}\Omega] = -13.32.$$

The results are compared in the following table

Parameter	*With r_d*	*With out r_d*
Z_i	1.5 MΩ	1.5 MΩ
A_V	– 12.08	– 13.32
Z_o	3.63 kΩ	4 kΩ

Observe that the results closely agree since $r_d = 10\ R_D$.

◆ 9.10 JFET COMMON SOURE AMPLIFIER USING SELF-BIAS CONFIGURATION

Figure 9.13(a) shows JFET common-source amplifier using self bias. Observe that, this circuit requires only one dc supply V_{DD} to establish the desired operating point. The dc voltage across R_S will serve as the biasing voltage between gate and source terminals.

$$\therefore \qquad V_{GS_Q} = -I_{DQ} R_S$$

The functions of C_1 and C_2 have already been explained in section 9.9. C_S is the source bypass capacitor. For ac operation it creates a short circuit across R_S, thus preventing ac negative feedback through R_S. Without C_S, the negative feedback through R_S reduces the voltage gain. Also C_S allows the dc bias current I_{DQ} through R_S, by acting as an open circuit.

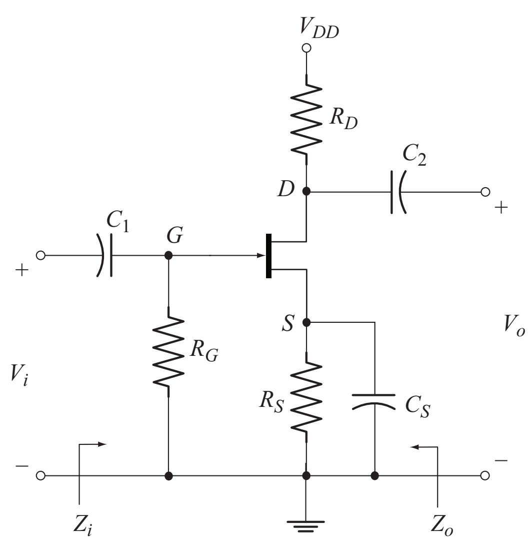


Fig. 9.13(a) JFET Common source amplifier using self bias configuration

The ac equivalent circuit is drawn in Fig 9.13(b) by short circuiting the capacitors C_1, C_2 and C_S and reducing V_{DD} to zero.

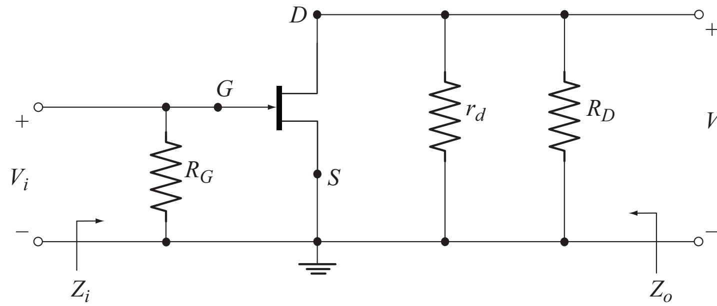


Fig. 9.13(b) *AC equivalent circuit of the amplifier of Fig 9.13(a)*

The ac equivalent circuit is redrawn in Fig. 9.14 by replacing the JFET with its small-signal ac model.

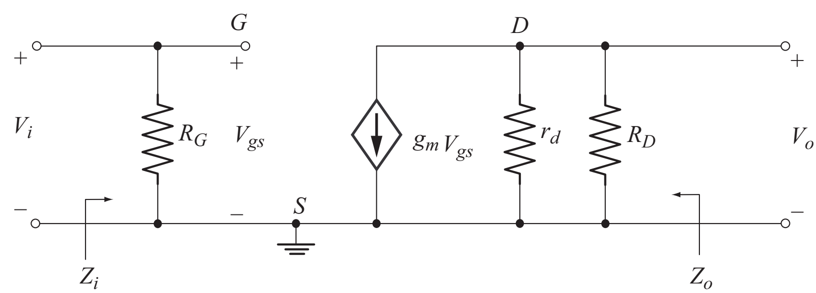


Fig. 9.14 *AC equivalent circuit using small signal model*

Observe that this circuit is same as that given in Fig. 9.10. Therefore all the results derived in section 9.9 can be readily applied to this circuit.

When the effect of r_d is considered.

$$Z_i = R_G \tag{9.26}$$

$$Z_o = r_d \,||\, R_D \tag{9.27}$$

$$A_V = -g_m\,[r_d \,||\, R_D\,] \tag{9.28}$$

For $r_d \geq 10\ R_D$

$$Z_o \approx R_D \tag{9.29}$$

$$A_V \approx -g_m R_D \tag{9.30}$$

The negative sign in the expression for A_V reveals that V_i and V_o are 180° out of phase.

Example 9.4

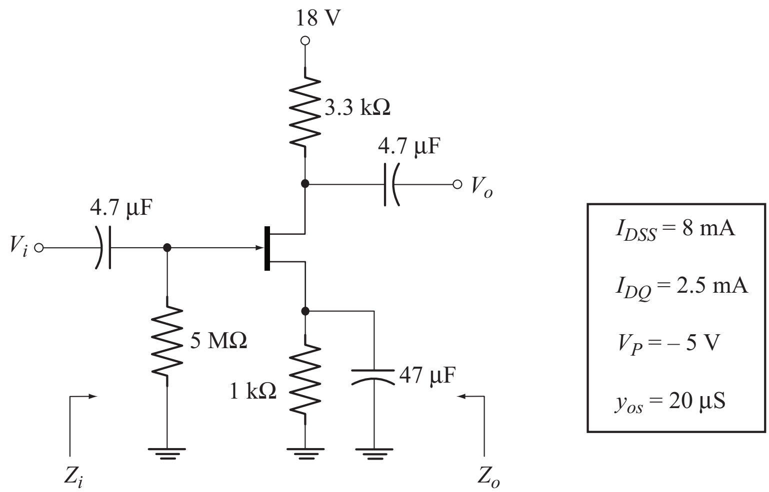
For the JFET amplifier shown below:

(a) Calculate Z_i and Z_o

(b) Calculage A_V

(c) Find V_o if V_i = 10 mV(p – p)

(d) Find Z_i, Z_o, A_V and V_o neglecting the effect of r_d.



Solution

Let us find g_m and r_d using the given data

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{2\,(8\text{ mA})}{5\text{V}} = 3.2\text{ mS}$$

$$V_{GS_Q} = -I_{DQ}R_S = -(2.5\text{ mA})\,(1\text{ k}\Omega) = -2.5\text{ V}$$

Now
$$g_m = 3.2\text{ mS}\left[1 - \frac{-2.5\text{ V}}{-5\text{ V}}\right] = 1.6\text{ mS}$$

$$r_d = \frac{1}{y_{o_s}} = \frac{1}{20\,\mu\text{S}} = 50\text{ k}\Omega$$

(a)
$$Z_i = R_G = 5\text{ M}\Omega$$

$$Z_o = r_d \,||\, R_D$$
$$= 50\text{ k}\Omega \,||\, 3.3\text{ k}\Omega = 3.09\text{ k}\Omega$$

(b)
$$A_V = -g_m\,[r_d \,||\, R_D] = -(1.6\text{ mS})\,(3.09\text{ k}\Omega) = -4.944$$

(c) $V_o = A_V V_i = (-4.944)(10 \text{ mV}) = -49.44 \text{ mV(p-p)}$

(d) $r_d = 50 \text{ k}\Omega$ and $10 R_D = 33 \text{ k}\Omega$

Since $r_d > 10 R_D$ we can neglect r_d. With r_d neglected, we get the following results.

$$Z_i = R_G = 5 \text{ M}\Omega$$

$$Z_o = R_D = 3.3 \text{ k}\Omega$$

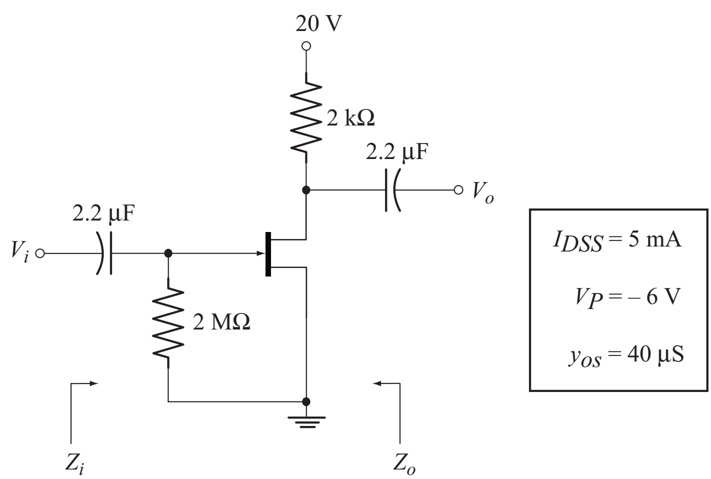
$$A_V = -g_m R_D = -(1.6 \text{ mS})(3.3 \text{ k}\Omega) = -5.28$$

$$V_o = (-5.28)(10 \text{ mV}) = -52.8 \text{ mV(p-p)}$$

Example 9.5

For the JFET amplifier shown below:

(a) Calculate Z_i and Z_o
(b) Calculate A_V
(c) Calculate V_o if $V_i = 50$ mV (rms)
(d) Calculate Z_i, Z_o, A_V and V_o, neglecting r_d



$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$V_{GS_Q} = I_{D_Q} R_S$$

Since $R_S = 0,\ V_{GS_Q} = 0$

$\therefore$ $g_m = g_{m_o}$

$$= \frac{2 I_{DSS}}{|V_p|} = \frac{(2)(5\text{ mA})}{6\text{ V}} = 1.67 \text{ mS}$$

$$r_d = \frac{1}{y_{o_S}} = \frac{1}{40\,\mu\text{S}} = 25\text{ k}\Omega$$

(a) $$Z_i = R_G = 2\text{ M}\Omega$$

$$Z_o = r_d \parallel R_D = 25\text{ k}\Omega \parallel 2\text{ k}\Omega = 1.85\text{ k}\Omega$$

(b) $$A_V = -g_m\,[r_d \parallel R_D] = -[1.67\text{ mS}]\,[1.85\text{ k}\Omega] = -3.08$$

(c) $$V_o = A_V\,V_i = (-3.08)\,(50\text{ mV}) = -154\text{ mV (rms)}$$

(d) $$r_d = 25\text{ k}\Omega \quad \text{and} \quad 10\,R_D = 20\text{ k}\Omega$$

Since $r_d > 10\,R_D$, we can neglect r_d.

Results with r_d neglected

$$Z_i = R_G = 2\text{ M}\Omega$$

$$Z_o = R_D = 2\text{ k}\Omega$$

$$A_V = -g_m\,R_D = -[1.67\text{ mS}]\,[2\text{ k}\Omega] = -3.34$$

$$V_o = AV_i = (-3.34)\,(50\text{ mV}) = -167\text{ mV (rms)}$$

9.11 JFET COMMON-SOURCE AMPLIFIER WITH UNBYPASSED R_S

Figure 9.15 shows the circuit of JFET common-source amplifier with unbypassed R_S. Since C_S is not present, ac feedback occurs through R_S. Note that the feedback voltage across R_S is in series with V_i and the feedback is negative.

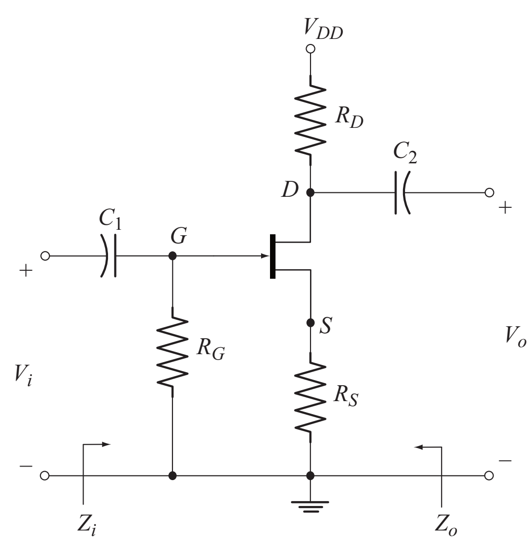


Fig. 9.15 JFET Common-source amplifier with unbypassed R_s

The ac equivalent circuit using the small signal ac model of JFET is drawn in Fig. 9.16

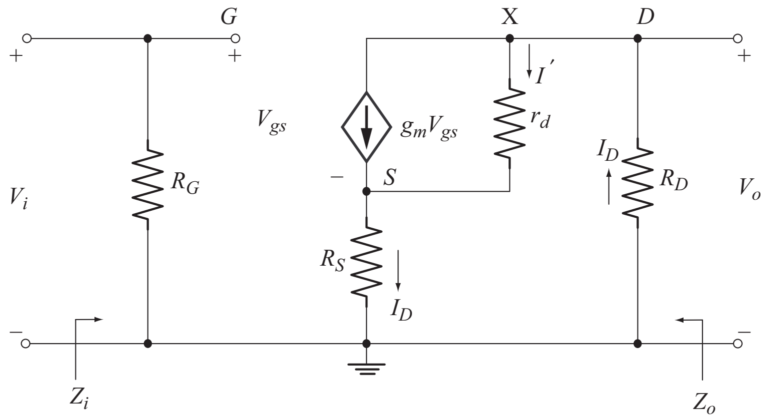


Fig. 9.16 Ac equivalent of the amplifier of Fig. 9.15

***Input Impedance* (z_i)**

From the input circuit of Fig. 9.16, we have

$$Z_i = R_G \tag{9.31}$$

***Voltage Gain* (A_V)**

$$A_V = \frac{V_o}{V_i} \tag{9.32}$$

$$V_o = -I_D R_D \tag{9.33}$$

To find I_D let us apply KCL at node X.

$$I_D = I' + g_m V_{gs} \tag{9.34}$$

Writing KVL to the path consisting of R_D, r_d and R_S we get

$$-I_D R_D - I' r_d - I_D R_S = 0$$

$$I' r_d = -I_D [R_D + R_S]$$

$$I' = -I_D \left[\frac{R_D + R_S}{r_d}\right] \tag{9.35}$$

Using Equation (9.35) in Equation (9.34) we get

$$I_D = -I_D \left[\frac{R_D + R_S}{r_d}\right] + g_m V_{gs} \tag{9.36}$$

To find V_{gs} let us apply kVL to the path consisting of V_i, V_{gs} and R_S we get

$$V_i - V_{gs} - I_D R_S = 0$$

$$V_{gs} = V_i - I_D R_S \tag{9.37}$$

Using this relation is Equation (9.36) we get

$$I_D = -I_D\left[\frac{R_D+R_S}{r_d}\right] + g_m\,[V_i - I_D\,R_S]$$

$$I_D\left[1+\frac{R_D+R_S}{r_d}+g_m R_S\right] = g_m V_i$$

$$I_D = \frac{g_m\ V_i}{1+g_m\ R_S+\dfrac{R_D+R_S}{r_d}} \tag{9.38}$$

Substituting Equation (9.38) in Equation (9.33) we get

$$V_o = -\frac{g_m\,R_D\,V_i}{1+g_m\,R_S+\dfrac{R_D+R_S}{r_d}}$$

Now $$A_V = \frac{V_o}{V_i} = -\frac{g_m\ R_D}{1+g_m\,R_S+\dfrac{R_D+R_S}{r_d}} \tag{9.39}$$

The negative sign in Equation (9.39) reveals that V_i and V_o differ in phase by 180°

Effect of Unbypassed R_S on Voltage Gain

If R_S is bypassed by C_S, it gets short circuited for ac operation. Therefore, substituting $R_S = 0$ in Equation (9.39),

we get $$A_V = -\frac{g_m\ R_D}{1+\dfrac{R_D}{r_d}}$$

$$= -g_m\left[\frac{R_D\ r_d}{R_D+r_d}\right]$$

$$= -g_m\,[r_d \,||\, R_D]$$

This result is consistent with Equation (9.28) derived in section 9.10.

Comparing this result with Equation (9.39), we find that, unbypassing of R_S results in the drastic reduction of voltage gain.

Expression for A_V neglecting the Effect of r_d

Consider Equation (9.39)

If $$r_d \geq 10\,(R_D + R_S)$$

or $$\left[\frac{R_D+R_S}{r_d}\right] \leq 0.1$$

$$1+g_m\,R_S + \left[\frac{R_D+R_S}{r_d}\right] \approx 1+g_m\,R_S$$

Now $$A_V \approx -\frac{g_m R_D}{1+g_m R_S} \tag{9.40}$$

Output Impedance (Z_o)

To develope the expression for Z_o we apply the following steps to the circuit of Fig. 9.16.

- The input voltage V_i is set to zero. As a result R_G gets short circuited.
- A voltage source V_o is connected between the output terminals.

The resulting circuit is shown in Fig. 9.17.

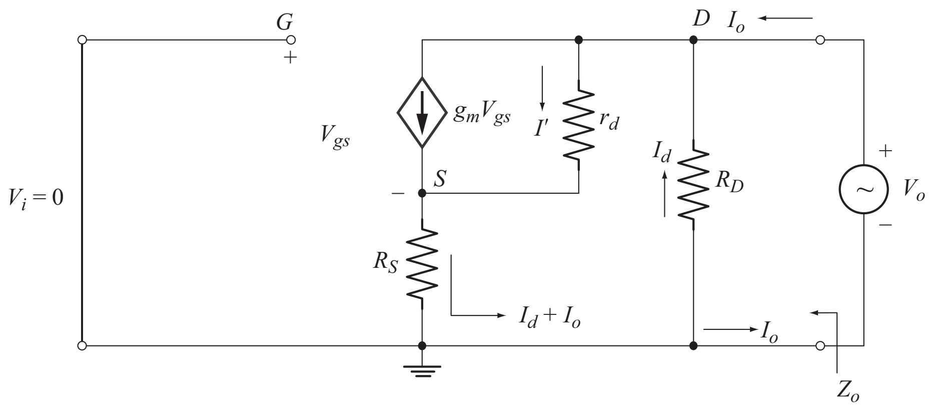


Fig. 9.17 Circuit to find Z_o

From Fig. 9.17, the output impedance is given by

$$Z_o = \frac{V_o}{I_o} \tag{9.41}$$

But $$V_o = -I_d R_D$$

using this relation in Equation (9.41) we get

$$Z_o = -\frac{I_D R_d}{I_o} \tag{9.42}$$

To express I_D in terms of I_o, let us apply KCL at the node D. We get

$$I_o + I_d - I' - g_m V_{gs} = 0 \tag{9.43}$$

Let us find I' and V_{gs} in terms of I_d and I_o. Applying KVL to the path consisting of R_D, r_d and R_S we have

$$-I_d R_D - I' r_d - (I_d + I_o) R_S = 0$$

$$-I' r_d = I_d [R_D + R_S] + I_o R_S$$

$$I' = -I_d\left[\frac{R_D + R_S}{r_d}\right] - I_o\left[\frac{R_S}{r_d}\right] \tag{9.44}$$

Applying KVL to the path consisting of V_i, V_{gs} and R_S we get

$$-V_{gs} - R_S\,[I_d + I_o] = 0$$

$$V_{gs} = -I_d R_S - I_o R_S \tag{9.45}$$

Substituting Equations (9.44) and (9.45) in Equation (9.43) we get

$$I_o + I_d + I_d\left[\frac{R_D + R_S}{r_d}\right] + I_o\left[\frac{R_S}{r_d}\right] + g_m R_s I_d + g_m I_o R_S = 0$$

$$I_o\left[1 + \frac{R_S}{r_d} + g_m R_S\right] = -I_d\left[1 + g_m R_S + \frac{R_D + R_S}{r_d}\right]$$

$$\frac{I_d}{I_o} = -\frac{1 + g_m R_S + \dfrac{R_S}{r_d}}{\left[1 + g_m R_S + \dfrac{R_D + R_S}{r_d}\right]} \tag{9.46}$$

Substituting this relation in Equation (9.42) we get

$$Z_o = \frac{\left[1 + g_m R_S + \dfrac{R_S}{r_d}\right] R_D}{\left[1 + g_m R_S + \dfrac{R_S}{r_d} + \dfrac{R_D}{r_d}\right]} \tag{9.47}$$

Expression for Z_o Neglecting the Effect of r_d

Consider Equation (9.47)

If $\quad r_d \geq 10\,R_D$

or $\quad \dfrac{R_D}{r_d} \leq 0.1$

then $\quad 1 + g_m R_S + \dfrac{R_S}{r_d} + \dfrac{R_D}{r_d} \approx 1 + g_m R_S + \dfrac{R_S}{r_d}$

Substituting this relation in Equation (9.47) we get

$$Z_o \approx R_D \tag{9.48}$$

Effect of Unbypassed R_S on Z_o

Observe from Equation (9.48) that, if $r_d \geq 10\ R_D$, unbypassing of R_S has no effect on Z_o. If $r_d < 10\ R_D$, Z_o is only slightly affected as given in Equation (9.47).

Important Characteristics of Common-source Configuration with Unbypassed R_S

Following are the important characteristics of common-source stage with unbypassed R_S.

- High input impedance
- Medium high output impedance
- Low voltage gain
- 180° phase shift betwen output and input voltages

Example 9.6

For the JFET amplifier shown below:

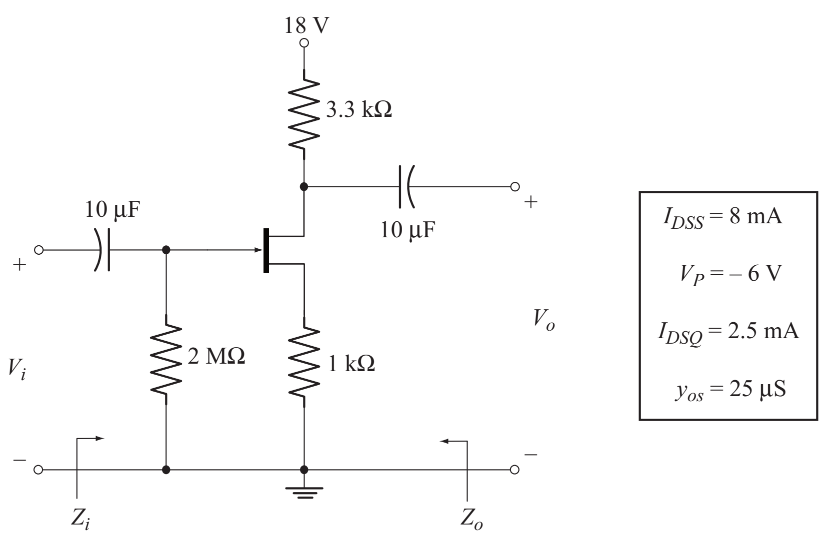
(a) Calculate Z_i and Z_o·

(b) Calculate A_V·

(c) Find V_o when $V_i = 50$ mV(p – p)·

(d) Calculate Z_i, Z_o, A_V and V_o neglecting the effect of r_d.

(e) Compare the results·



Solution

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_P}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|}$$

$$= \frac{2\left(8\,\text{mA}\right)}{6\,\text{V}} = 2.67 \text{ mS}$$

$$V_{GS_Q} = -I_{D_Q} R_S$$

$$= -(2.5 \text{ mA})(1 \text{ k}\Omega)$$

$$= -2.5 \text{ V}$$

Now $$g_m = 2.67 \text{ mS}\left[1 - \frac{-2.5\,\text{V}}{-6\,\text{V}}\right]$$

$$= 1.55 \text{ mS}$$

$$r_d = \frac{1}{y_{o_S}}$$

$$= \frac{1}{25\,\mu S} = 40 \text{ k}\Omega$$

(a) $$Z_i = R_G = 2 \text{ M}\Omega$$

$$Z_o = \frac{\left[1 + g_m R_S + \frac{R_S}{r_d}\right] R_D}{\left[1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d}\right]}$$

$$1 + g_m R_S + \frac{R_S}{r_d} = 1 + (1.55 \text{ mS})(1 \text{ k}\Omega) + \frac{1\,\text{k}\Omega}{40\,\text{k}\Omega} = 2.575$$

$$1 + g_m R_S + \frac{R_S}{r_d} + \frac{R_D}{r_d} = 2.575 + \frac{3.3\,\text{k}\Omega}{40\,\text{k}\Omega} = 2.657$$

Now $$Z_o = \frac{(2.575)(3.3\,\text{k}\Omega)}{2.657} = 3.198 \text{ k}\Omega$$

(b) $$A_V = -\frac{g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

$$= -\frac{(1.55\,\text{mS})(3.3\,\text{k}\Omega)}{1 + (1.55\,\text{mS})(1\,\text{k}\Omega) + \left(\frac{3.3\,\text{k}\Omega + 1\,\text{k}\Omega}{40\,\text{k}\Omega}\right)}$$

$$= -1.924$$

(c)
$$V_o = A_V V_i$$
$$= (-1.924)\,(50 \text{ mV})$$
$$= -96.2 \text{ mV}_{(p-p)}$$

(d) To find Z_i, Z_o, A_V and V_o neglecting the effect of r_d.

$$Z_i = R_G = 2 \text{ M}\Omega \quad \text{(Same as before)}$$
$$r_d = 40 \text{ k}\Omega \quad \text{and} \quad 10\,R_D = 33 \text{ k}\Omega$$

Since $r_d > 10\,R_D$, we can use Equation (9.48) to calculate Z_o.

$$Z_o = R_D = 3.3 \text{ k}\Omega$$

In order to use Equation (9.40) to find A_V, the required condition is

$$r_d \geq 10\,(R_D + R_S)$$
$$10\,(R_D + R_S) = 10\,(3.3 \text{ k}\Omega + 1\text{k}\Omega) = 43 \text{ k}\Omega$$
$$\text{But} \quad r_d = 40 \text{ k}\Omega$$

Though the above condition is not satisfied, r_d is nearly equal to 10 $(R_D + R_S)$. Hence we can use Equation (9.40).

$$A_v = -\frac{g_m R_D}{1+g_m R_S}$$
$$= -\frac{(1.55\,\text{mS})(3.3\,\text{k}\Omega)}{1+(1.55\,\text{mS})(1\,\text{k}\Omega)} = -2$$

The results are tabulated in the following table

Parameter	***Considering the effect of r_d***	***Neglecting the effect of r_d***
Z_i	2 MΩ	2 MΩ
Z_o	3.198 kΩ	3.3 kΩ
A_V	– 1.924	– 2

Observe that, the results are almost consistent since the approximating conditions are satisfied.

◆ 9.12 JFET COMMON-SOURCE AMPLIFIER USING VOLTAGE DIVIDER CONFIGURATION

Figure 9.18 shows the circuit of JFET common-source amplifier using voltage divider configuration.

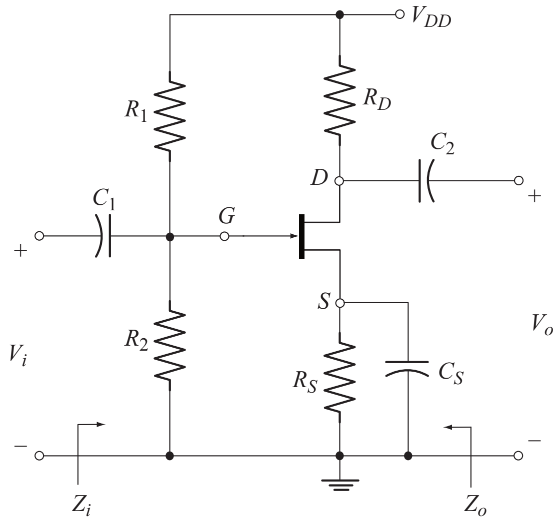


Fig. 9.18 JFET common-source amplifier using voltage divider configuration

The ac equivalent circuit is drawn in Fig. 9.19 by reducing V_{DD} to zero and short circuiting the capacitors C_1, C_2 and C_S. Observe that, the top end of R_1 goes to ground. As a result R_1 and R_2 will come in parallel between gate and the source.

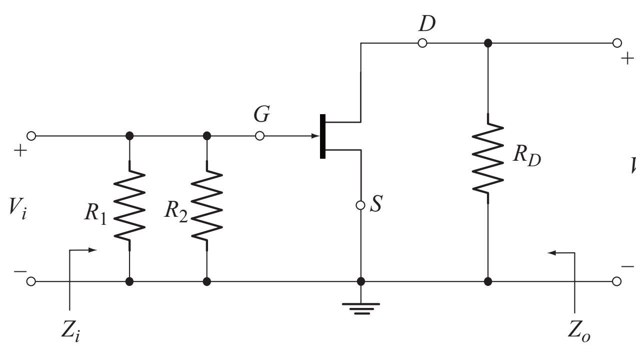


Fig. 9.19 Ac equivalent Circuit of the amplifier of Fig. 9.18

The ac equivalent circuit is redrawn in Fig. 9.20 by replacing JFET with its small signal ac model.

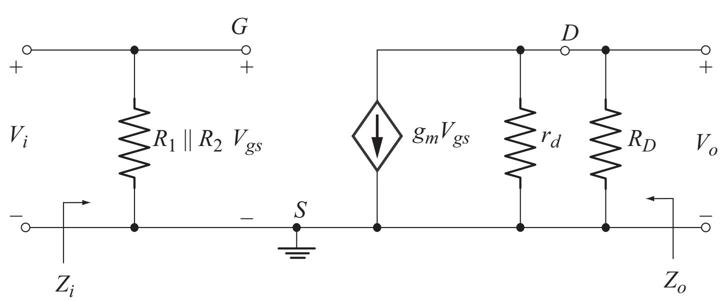


Fig. 9.20 Ac equivalent circuit redrawn using small-signal ac model of JFET

Note that the ac equivalent circuit of Fig. 9.20 is exactly the same as that of JFET amplifier using fixed bias configuration, shown in Fig. 9.10. Hence the results derived in section 9.9 can be readily applied to this circuit. The only exception is that, R_G should be replaced by $R_1 \parallel R_2$. Let us rewrite the results for reference.

When the effect of r_d is included

From Equations (9.18), (9.19) and (9.23) we have

$$Z_i = R_1 \parallel R_2 \tag{9.49}$$

$$Z_o = r_d \parallel R_D \tag{9.50}$$

$$A_V = -g_m [r_d \parallel R_D] \tag{9.51}$$

When the effect of r_d is neglected [For $r_d \geq 10\, R_D$]

From Equations (9.24) and (9.25) we have

$$Z_o \approx R_D \tag{9.52}$$

$$A_V \approx -g_m R_D \tag{9.53}$$

$$Z_i = R_1 \parallel R_2 \tag{9.54}$$

◆ 9.13 JFET COMMON-SOURCE AMPLIFIER USING VOLTAGE DIVIDER CONFIGURATION, WITH UNBYPASSED R_S

Figure 9.21 shows the circuit of JFET common-source amplifier using voltage divider configuration, with unbypassed R_S.

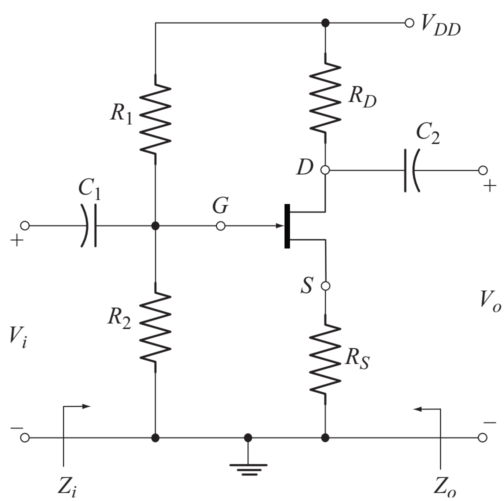


Fig. 9.21 JFET Voltage-divider configuration with unbypassed R_S

The ac equivalent circuit is drawn in Fig. 9.22 by reducing V_{DD} to zero, short circuiting capacitors C_1 and C_2 and replacing JFET with its small signal ac model.

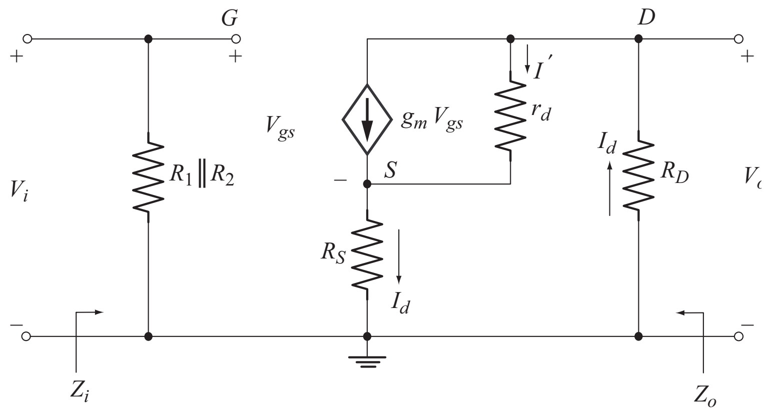


Fig. 9.22 Ac equivalent circuit of the JFET amplifier of Fig. 9.21

The ac equivalent circuit of Fig. 9.22 is exactly the same as that of JFET self bias configuration with unbypassed R_S, shown in Fig 9.16. Hence the results derived in section 9.11 can be readily applied to this circuit. The only exception is that, R_G should be replaced by $R_1 \| R_2$. Let us recall the results for reference.

When the effect of r_d is included

From Equations (9.31), (9.39) and (9.47) we have

$$Z_i = R_1 \| R_2 \tag{9.55}$$

$$A_V = -\frac{g_m R_D}{1 + g_m R_S + \dfrac{R_D + R_S}{r_d}} \tag{9.56}$$

$$Z_o = \frac{\left[1 + g_m R_S + \dfrac{R_S}{r_d}\right] R_D}{\left[1 + g_m R_S + \dfrac{R_S}{r_d} + \dfrac{R_D}{r_d}\right]} \tag{9.57}$$

When the effect of r_d is neglected

Form Equations (9.40) and (9.48) we have

$$A_V = -\frac{g_m R_D}{1 + g_m R_S} \quad [\text{when } r_d \geq 10\,[R_D + R_S]] \tag{9.58}$$

$$Z_o = R_D \quad [\text{when } r_d \geq 10\,R_D] \tag{9.59}$$

$$Z_i = R_1 \| R_2 \tag{9.60}$$

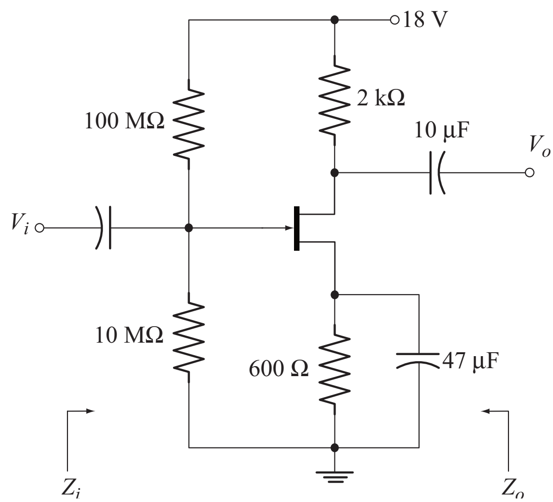
Example 9.7

For the JFET amplifier shown below:

(a) Calculate Z_i and Z_o

(b) Calculate A_V

(c) Find V_o if $V_i = 25$ mV (rms)



$I_{DSS} = 12$ mA

$V_P = -3$ V

$y_{os} = 10$ μS

$V_{GSQ} = -1$ V

Solution

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_P}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|}$$

$$= \frac{2(12\,\text{mA})}{3\,\text{V}} = 8\text{ mS}$$

$$g_m = 8\text{ mS}\left[1 - \frac{-1\,\text{V}}{-3\,\text{V}}\right] = 5.33\text{ mS}$$

$$r_d = \frac{1}{y_{o_s}} = \frac{1}{10\,\mu\text{S}} = 100\text{ k}\Omega$$

(a) $$Z_i = R_1 \parallel R_2 = 100\text{ M}\Omega \parallel 10\text{ M}\Omega = 9.09\text{ M}\Omega$$

$$Z_o = r_d \parallel R_D = 100\text{ k}\Omega \parallel 2\text{ k}\Omega = 1.96\text{ k}\Omega$$

(b) $$A_V = -g_m\,[r_d \parallel R_D]$$

$$= -[5.33\text{ mS}]\,[1.96\text{ k}\Omega] = -10.44$$

(c)
$$V_o = A_V V_i$$
$$= [-10.44]\,[25\text{ mV}] = -0.261\text{ V (rms)}$$

Example 9.8

Repeat example 9.7 with source bypass capacitor C_S removed and compare the results.

Solution

$$\left.\begin{aligned} g_m &= 5.33\text{ mS} \\ r_d &= 100\ \text{k}\Omega \end{aligned}\right\}\ \text{As calculated in example 9.7}$$

(a)
$$Z_i = R_1 \parallel R_2 = 9.09\text{ k}\Omega$$
$$10\,R_D = 10\,(2\text{ k}\Omega) = 20\text{ k}\Omega$$

Note that $r_d > 10\,R_D$

$\therefore$ $Z_o = R_D$

$$Z_o = 2\text{ k}\Omega$$

(b)
$$10\,(R_S + R_D) = 10\,(600\ \Omega + 2\text{ k}\Omega) = 26\text{ k}\Omega$$

Note that $r_d > 10\,(R_S + R_D)$

$$\therefore \quad A_V = -\frac{g_m R_D}{1 + g_m R_S}$$
$$= -\frac{(5.33\text{mS})(2\text{kW})}{1 + (5.33\text{ mS})(600\text{ W})}$$
$$= -2.53$$

(c)
$$V_o = A_V V_i$$
$$= (-2.53)\,(25\text{ mV}) = -63.25\text{ mV (rms)}$$

The results of examples 9.7 and 9.8 are compared in the following table.

Parameter	*With bypassed R_S*	*With unbypassed R_S*	*Remarks*
Z_i	9.09 kΩ	9.09 kΩ	Not affected
Z_o	1.96 kΩ	2 kΩ	Slightly affected
A_V	– 10.44	– 2.53	Considerable reduction in voltage gain

Inference: Unbypassing of R_S in common-source configuration results in considerable reduction in voltage gain.

Example 9.9

Repeat example 9.8 taking $r_d = 20\ \text{k}\Omega$ keeping all other data remain unchanged.

Solution

$$g_m = 5.33\ \text{mS} \quad \text{[as given in example 9.9]}$$

$$r_d = 20\ \text{k}\Omega$$

(a)
$$Z_i = R_1 \parallel R_2 = 9.09\ \text{k}\Omega$$

$$10\ R_D = (10)\ (2\ \text{k}\Omega) = 20\ \text{k}\Omega$$

The condition, $r_d \geq 10\ R_D$ is satisfied

$$\therefore \quad Z_o = R_D = 2\ \text{k}\Omega$$

(b)
$$10\ (R_S + R_D) = 10\ (600\ \Omega + 2\ \text{k}\Omega) = 26\ \text{k}\Omega$$

Note that, $r_d < 10\ (R_S + R_D)$

$$\therefore \quad A_V = -\frac{g_m\ R_D}{1 + g_m\ R_S + \dfrac{R_D + R_S}{r_d}}$$

$$= -\frac{(5.33\ \text{mS})(2\,\text{k}\Omega)}{1 + (5.33\,\text{mS})(600\,\Omega) + \dfrac{2.6\,\text{k}\Omega}{20\,\text{k}\Omega}} = -2.46$$

◆ 9.14 JFET SOURCE-FOLLOWER [COMMON-DRAIN CONFIGURATION]

Figure 9.23 shows the circuit of JFET source-follower configuration. This circuit is analogous to BJT emitter-follower configuration.

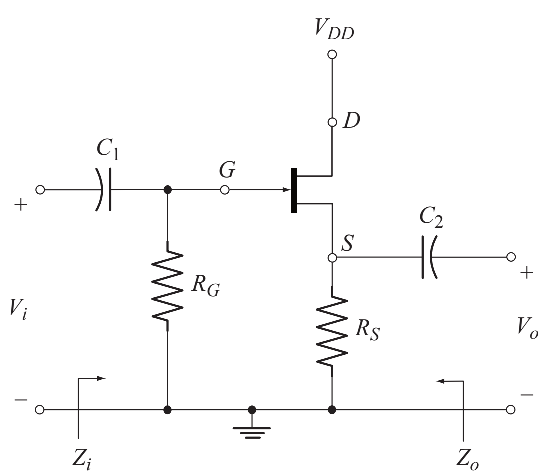


Fig. 9.23 JFET source-follower configuration

The ac equivalent circuit is drawn in Fig. 9.24, by reducing V_{DD} to zero and replacing C_1 and C_2 by their short circuit equivalents.

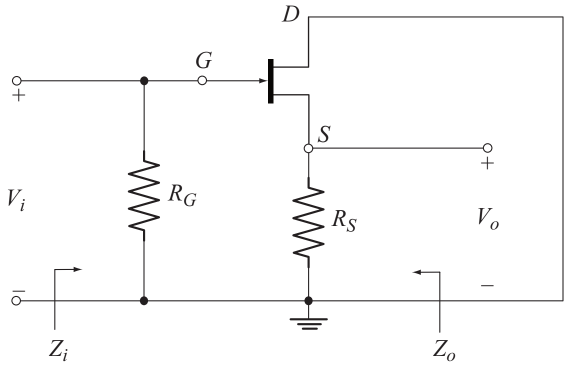


Fig. 9.24 AC equivalent Circuit of JFET source-follower of Fig. 9.23

Observe from Fig. 9.24 that, the ac input signal V_i, is applied between gate and ground (drain) and the ac output signal V_o, is taken between source and ground (drain). Since the drain terminal is common to input and output signals, the circuit is called common drain configuration.

The circuit has a voltage gain of approximately unity. Therefore $V_o \approx V_i$. Note that the output voltage (source voltage) follows the input voltage (gate voltage). Hence this circuit is also called the source-follower.

The ac equivalent circuit is redrawn in Fig. 9.25 using the JFET small-signal ac model.

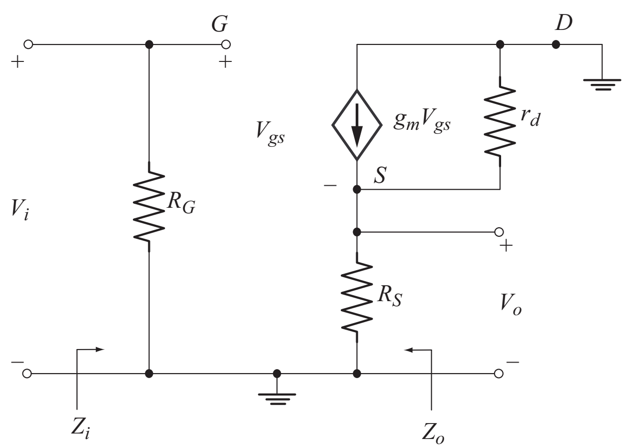


Fig. 9.25 Ac equivalent circuit of source-follower using JFET small-signal ac model

The ac equivalent circuit of Fig. 9.25 is redrawn in Fig. 9.26 for the convenience of analysis. This Circuit is written based on the following details available in the circuit of Fig. 9.25.

- The current $g_m V_{gs}$ flows from drain to source.
- Both r_d and R_S appear between source and drain terminals.

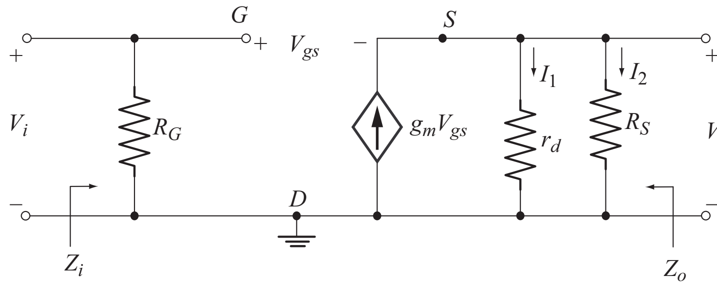


Fig. 9.26 Ac equivalent circuit of Fig. 9.25 redrawn

Input Impedance **[Z_i]**

From the input circuit of Fig. 9.26, we have

$$Z_i = R_G \tag{9.61}$$

Voltage Gain **[A_V]**

Applying KCL at the source node of Fig. 9.26 we have

$$g_m V_{gs} = \frac{V_o}{r_d} + \frac{V_o}{R_S}$$

$$g_m V_{gs} = V_o\left[\frac{1}{r_d} + \frac{1}{R_S}\right]$$

$$g_m V_{gs} = V_o\left[\frac{R_S + r_d}{R_S\ r_d}\right]$$

$$V_o = g_m V_{gs}\left[\frac{R_S\ r_d}{R_S + r_d}\right]$$

or

$$V_o = g_m V_{gs} [r_d \,||\, R_S] \tag{9.62}$$

Applying KVL to the path consisting of V_i, V_{gs} and V_o we get

$$V_i - V_{gs} - V_o = 0 \tag{9.63}$$

or

$$V_{gs} = V_i - V_o \tag{9.64}$$

Using this relation in Equation (9.62) we get

$$V_o = g_m [V_i - V_o] [r_d \,||\, R_S]$$

$$V_o = g_m V_i [r_d \,||\, R_S] - g_m V_o [r_d \,||\, R_S]$$

$$V_o \big[1 + g_m [r_d \,||\, R_S]\big] = g_m V_i [r_d \,||\, R_S]$$

Now the voltage gain is

$$A_V = \frac{V_o}{V_i} = \frac{g_m\,[r_d \,\|\, R_S]}{1 + g_m\,[r_d \,\|\, R_S]} \tag{9.65}$$

Positive sign for A_V reveals that, V_o and V_i are in phase. Note that source-follower configuration is a non inverting amplifier.

Output Impedance **[Z_o]**

To find the output impedance, we apply the following steps to the circuit of Fig. 9.26:

- The input voltage V_i is set to zero. As a result R_G gets short Circuited
- A voltage source V_o is connected between the output terminals.

The resulting circuit is shown in Fig. 9.27.

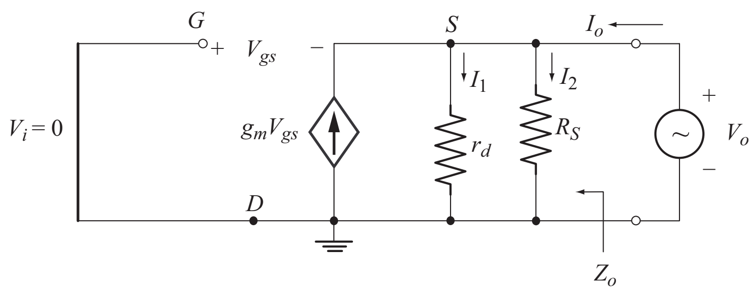


Fig. 9.27 Circuit to Find Z_o

From Fig. 9.27, the output impedance is given by

$$Z_o = \frac{V_o}{I_o} \tag{9.66}$$

Applying KCL at the node *S*, we have

$$I_o + g_m V_{gs} - I_1 - I_2 = 0 \tag{9.67}$$

$$I_1 = \frac{V_o}{r_d} \quad \text{and} \quad I_2 = \frac{V_o}{R_S}$$

Using these relations in Equation (9.67) we get

$$I_o + g_m V_{gs} - \frac{V_o}{r_d} - \frac{V_o}{R_S} = 0 \tag{9.68}$$

Let us eliminate V_{gs} by applying KVL to the path consisting of V_i, V_{gs} and V_o. We get

$$V_i - V_{gs} - V_o = 0$$

Since $V_i = 0$, we have

$$V_{gs} = -V_o \tag{9.69}$$

Using this relation in Equation (9.68) we get

$$I_o - g_m V_o - \frac{V_o}{r_d} - \frac{V_o}{R_S} = 0$$

$$I_o = V_o\left[g_m + \frac{1}{r_d} + \frac{1}{R_S}\right]$$

Now
$$Z_o = \frac{V_o}{I_o} = \frac{1}{g_m + \frac{1}{r_d} + \frac{1}{R_S}} \tag{9.70}$$

or
$$Z_o = \frac{1}{\frac{1}{\left[\frac{1}{g_m}\right]} + \frac{1}{r_d} + \frac{1}{R_S}} \tag{9.71}$$

In Equation (9.71), Z_o can be interpreted as the parallel combination of $\frac{1}{g_m}$, r_d and R_S.

$$\therefore \qquad Z_o = \frac{1}{g_m} \parallel r_d \parallel R_S \tag{9.72}$$

Expression for A_v Neglecting the effect of r_d

For $r_d \geq 10\, R_S$

$$r_d \parallel R_S \approx R_S$$

From Equation (9.65) we get

$$A_V = \frac{V_o}{V_i} \approx \frac{g_m R_S}{1 + g_m R_S} \tag{9.73}$$

Further, if $g_m R_S \gg 1$, then from Equation (9.73) we get

$$A_V \approx \frac{g_m R_S}{g_m R_S} = 1$$

Note that the voltage gain under this condition is unity which justifies the name source-follower.

Expression for Z_o neglecting the effect of r_d

For $r_d \geq 10\, R_S$

$$r_d \parallel R_S \approx R_S$$

Using this condition in Equation (9.72) we get

$$Z_o \approx \frac{1}{g_m} \parallel R_S \tag{9.74}$$

Usually $g_m R_S \gg 1$ or $\frac{1}{g_m} \ll R_S$

As a result $Z_o \approx \frac{1}{g_m}$

Observe that source-follower has low output impedance relative to that of common-source and common-gate configurations.

Important characteristics of source-follower

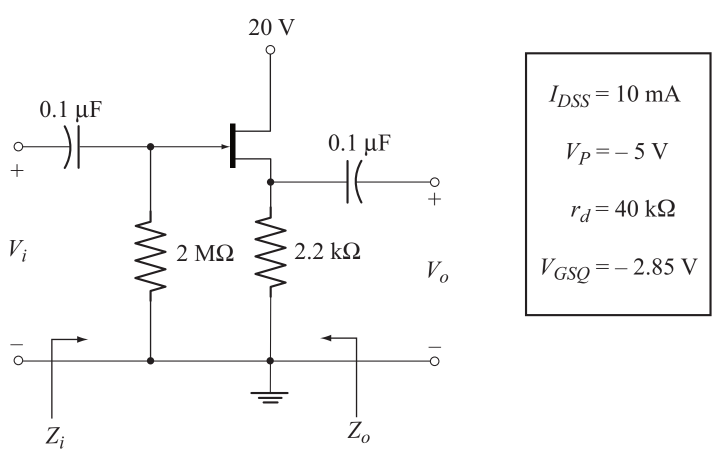
Following are the important characteristics of source-follower

- Approximately unity voltage gain
- High input impedance
- Relatively low output impedance
- Input and output voltages are in phase.

Example 9.10

For the JFET common drain configuration shown below:

(a) Calculate Z_i and Z_o
(b) Calculate A_V
(c) Find V_o if $V_i = 20$ mV(p – p)
(d) Find Z_i, Z_o, A_V and V_o neglecting the effect of r_d.



Solution

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_P}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{2\,(10\text{ mA})}{5\text{ V}} = 4\text{ mS}$$

$$g_m = 4\text{ mS}\left[1 - \frac{-2.85\text{ V}}{-5\text{ V}}\right] = 1.72\text{ mS}$$

(a)
$$Z_i = R_G = 2\ \text{M}\Omega$$
$$Z_o = \frac{1}{g_m} \parallel r_d \parallel R_S$$
$$\frac{1}{g_m} = \frac{1}{1.72\ \text{mS}} = 581.39\ \Omega$$
$$Z_o = 581.39\Omega \parallel 40\ \text{k}\Omega \parallel 2.2\ \text{k}\Omega = 454.63\Omega$$

(b)
$$A_V = \frac{g_m\left[r_d \parallel R_S\right]}{1 + g_m\left[r_d \parallel R_S\right]}$$
$$= \frac{\left[1.72\,\text{mS}\right]\left[40\,\text{k}\Omega \parallel 2.2\,\text{k}\Omega\right]}{1+\left[1.72\,\text{mS}\right]\left[40\,\text{k}\Omega \parallel 2.2\,\text{k}\Omega\right]} = 0.7818$$

(c)
$$V_o = A_V V_i = [0.7818]\ [20\ \text{mV}] = 15.636\ \text{mV(p – p)}$$

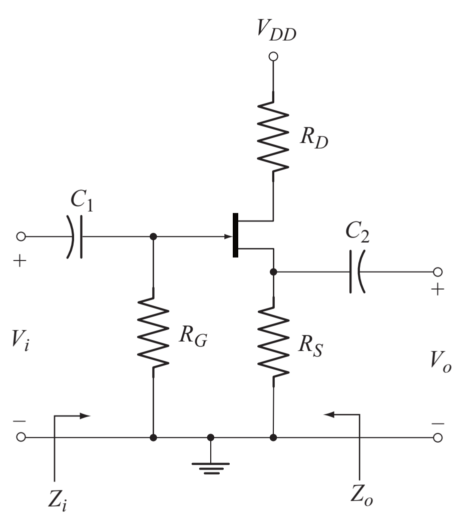
(d)
$$r_d = 40\ \text{k}\Omega \quad \text{and} \quad 10\ R_S = 22\ \text{k}\Omega$$

Since $r_d > 10\ R_S$, we can neglect the effect of r_d.

$$Z_i = 2\ \text{M}\Omega$$
$$Z_o = \frac{1}{g_m} \parallel R_S = 581.39\ \Omega \parallel 2.2\ \text{k}\Omega = 459.86\ \Omega$$
$$A_V = \frac{g_m\ R_S}{1 + g_m\ R_S} = \frac{(1.72\,\text{mS})(2.2\,\text{k}\Omega)}{1+(1.72\,\text{mS})(2.2\,\text{k}\Omega)} = 0.79$$
$$V_o = A_V V_i = (0.79)\ (20\ \text{mV}) = 15.8\ \text{mV(p – p)}$$

Example 9.11

For the source-follower with the drain circuit resistor R_D shown below, derive the expressions for Z_i, A_V and Z_o.



Solution

In the ac equivalent circuit, R_D comes between drain and ground. The ac equivalent circuit is drawn in Fig (A), using the circuit of Fig. 9.25.

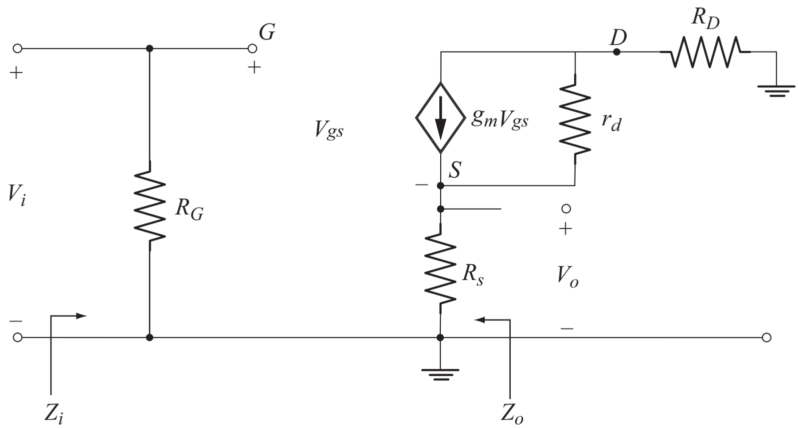
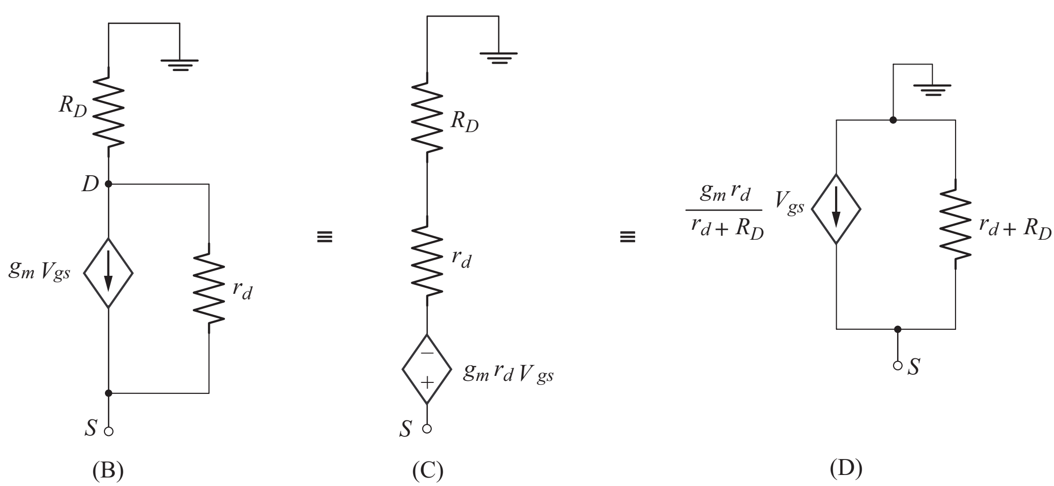


Fig. (A)

A part of the output circuit is shown in Fig. (B).



The current source $g_m V_{gs}$ in parallel with r_d in Fig. B is converted into its equivalent voltage source as shown Fig. (C). The voltage source is converted back into its equivalent current source in Fig. (D).

Let $$g'_m = \frac{g_m r_d}{r_d + R_D} \tag{A}$$

and $$r_d' = r_d + R_D \tag{B}$$

The ac equivalent circuit is redrawn in Fig. (E) after substituting the circuit of Fig (D) in Fig. (A).

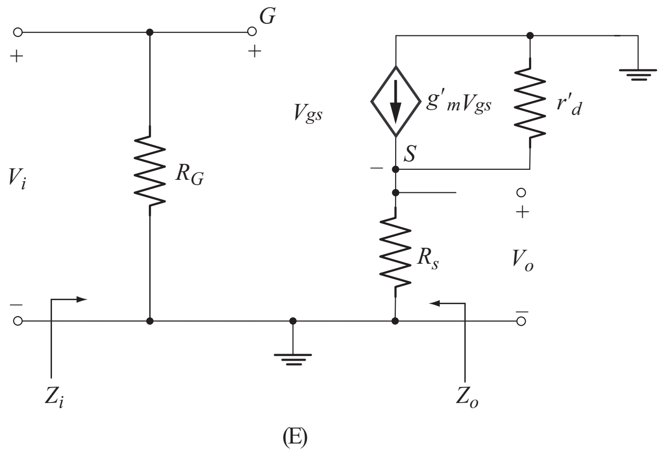


Fig. (E)

The circuit of Fig. (E) has the same format as that of Fig. 9.25. Proceeding on the same lines given in section 9.14 we can arrive at the following results.

$$Z_i = R_G \tag{C}$$

$$A_V = \frac{V_o}{V_i} = \frac{g'_m\left[r'_d \| R_S\right]}{1+g'_m\left[r'_d \| R_S\right]} \tag{D}$$

$$Z_o = \frac{1}{g'_m} \| r'_d \| R_S \tag{E}$$

Example 9.12

The following data are available for the circuit of example 9.11:

$V_{DD} = 20$ V	$R_D = 3.3$ kΩ	$R_G = 12$ MΩ
$R_S = 3.3$ kΩ	$I_{DSS} = 6$ mA	$V_p = -6$ V
$r_d = 30$ kΩ	$V_{GS_Q} = -3.8$ V	

Calculate the values of Z_i, A_V and Z_o.

Solution

$$Z_i = R_G = 12 \text{ M}\Omega$$

$$r'_d = r_d + R_D = 30 \text{ k}\Omega + 3.3 \text{ k}\Omega = 33.3 \text{ k}\Omega$$

$$g'_m = \frac{g_m\, r_d}{r_d + R_D}$$

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_m = \frac{2\,I_{DSS}}{|V_P|} = \frac{2\ (6\ \text{mA})}{6\ \text{V}} = 2\ \text{mS}$$

$$g_{m_o} = 2\ \text{mS}\left[1 - \frac{-3.8\,\text{V}}{-6\,\text{V}}\right] = 0.733\ \text{mS}$$

Now $$g'_m = \frac{[0.733\,\text{mS}][30\,\text{k}\Omega]}{[30\,\text{k}\Omega + 3.3\,\text{k}\Omega]} = 0.66\ \text{mS}$$

$$A_V = \frac{g'_m\,[r'_d \parallel R_S]}{1 + g'_m\,[r'_d \parallel R_S]}$$

$$= \frac{0.66\,\text{mS}\,[33.3\,\text{k}\Omega \parallel 3.3\,\text{k}\Omega]}{1 + 0.66\,\text{mS}\,[33.3\,\text{k}\Omega \parallel 3.3\,\text{k}\Omega]}$$

$$= 0.66$$

$$Z_o = \frac{1}{g'_m} \parallel r'_d \parallel R_S$$

$$\frac{1}{g'_m} = \frac{1}{0.66\,\text{mS}} = 1.515\ \text{k}\Omega$$

$$\therefore\quad Z_o = 1.515\ \text{k}\Omega \parallel 33.3\ \text{k}\Omega \parallel 3.3\ \text{k}\Omega$$

$$= 1\ \text{k}\Omega$$

9.15 JFET COMMON-GATE CONFIGURATION

JFET common-gate configuration is analogous to the BJT common-base configuration. Figure 9.28 shows the circuit of JFET common-gate configuration. An alternate representation of this circuit is also shown in Fig. 9.29.

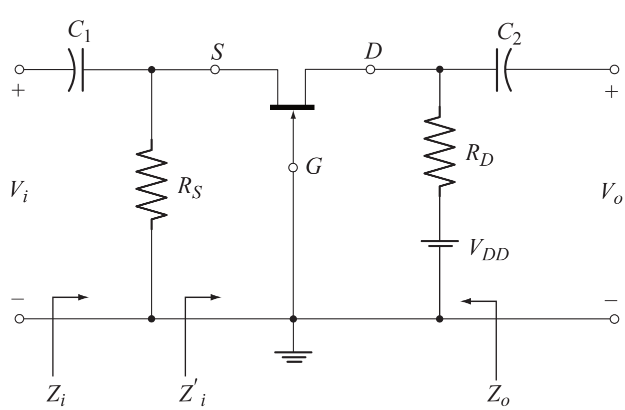


Fig. 9.28 JFET common-gate configuration

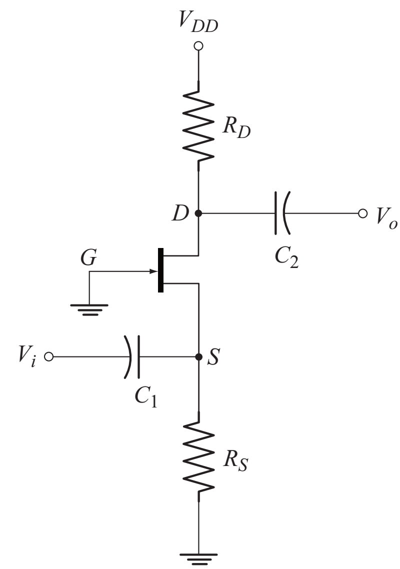


Fig. 9.29 Alternate representation of the circuit of Fig. 9.28

The ac equivalent circuit of Fig. 9.28 is drawn in Fig. 9.30 by replacing C_1 and C_2 with their short circuit equivalents and reducing V_{DD} to zero.

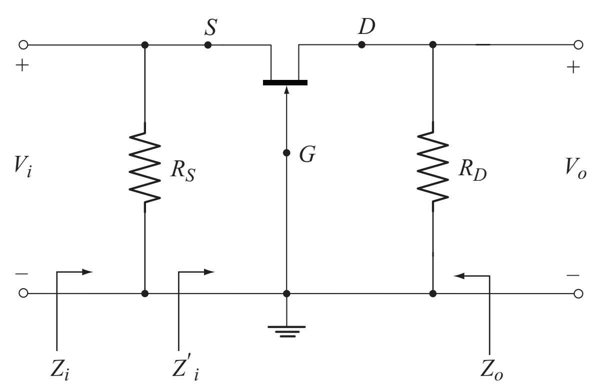


Fig. 9.30 Ac equivalent circuit of common-gate configuration

Observe form Fig. 9.30 that, the gate terminal is common for measuring both V_o and V_i. Hence the name common-gate configuration.

The ac equivalent is redrawn in Fig. 9.31 by replacing JFET with its small-signal ac model. Let us make the following important observations from the circuit of Fig. 9.31.

- R_S appears between source and gate. As a result, the open circuit condition between gate and source has been obviously lost.
- R_D appears between drain and gate.
- The current source $g_m V_{gs}$ and the FET output resistance r_d, will both appear between the drain and source terminals. The direction of $g_m V_{gs}$ is from drain to source, as usual.

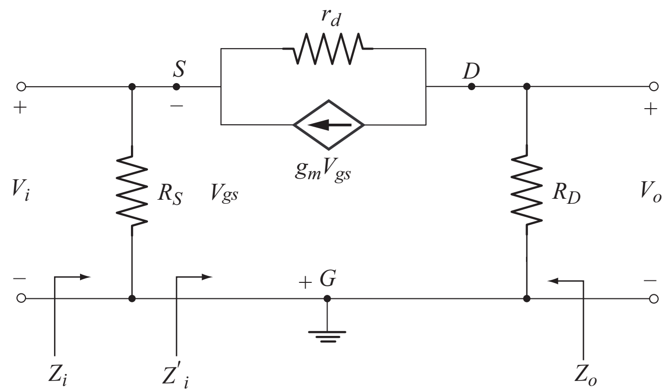


Fig. 9.31 AC equivalent circuit using JFET small signal model

Input Impedance [Z_i]

Form Fig. 9.31, the input impedance, Z_i is given by the parallel combination of R_S and Z'_i.

$$\text{i.e.,} \qquad Z_i = R_S \parallel Z'_i \tag{9.75}$$

where Z'_i is the impedance seen looking into source and gate terminals, excluding R_S.

To find Z'_i

The circuit between source and gate of Fig. 9.31 omitting R_S is redrawn in the convenient form in Fig. 9.32, to find Z'_i.

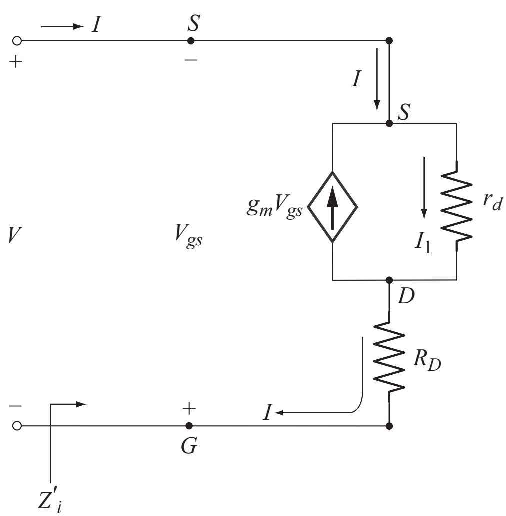


Fig. 9.32 Circuit to find Z_i'

Form Fig. 9.32, we find that

$$Z'_i = \frac{V}{I} \tag{9.76}$$

Also $$V = -V_{gs} \tag{9.77}$$

Applying KCL at the node S, we have

$$I + g_m V_{gs} = I_1 \tag{9.78}$$

Substituting Equation (9.77) in (9.78), we get

$$I - g_m V = I_1 \tag{9.79}$$

To express I_1 interms of V and I, let us apply KVL to the path consisting of V, r_d and R_D. We get

$$V - I_1 r_d - I R_D = 0$$

$$I_1 = \frac{V}{r_d} - \frac{R_D}{r_d} I \tag{9.80}$$

Using Equation (9.80) in Equation (9.79), we get

$$I - g_m V = \frac{V}{r_d} - \frac{R_D}{r_d} I$$

$$I\left[1 + \frac{R_D}{r_d}\right] = V\left[g_m + \frac{1}{r_d}\right]$$

$$Z_i' = \frac{V}{I} = \frac{\left[1 + \frac{R_D}{r_d}\right]}{\left[g_m + \frac{1}{r_d}\right]} \tag{9.81}$$

or $$Z_i' = \frac{V}{I} = \frac{r_d + R_D}{1 + g_m r_d} \tag{9.82}$$

Using this relation in Equation (9.75), we get

$$Z_i = R_S \,||\, \left[\frac{r_d + R_D}{1 + g_m r_d}\right] \tag{9.83}$$

Expression for Voltage Gain [A_v]

$$A_V = \frac{V_o}{V_i} \tag{9.84}$$

From Fig. 9.31 we have

$$V_i = -V_{gs} \tag{9.85}$$

Combining Equations (9.77) and (9.85) we get

$$V_i = V \tag{9.86}$$

Observe from Fig. 9.31 that, V_o is measured across R_D connected between drain and ground i.e., V_o is the voltage across R_D.

From Fig. 9.32,

$$V_o = I R_D \tag{9.87}$$

From Equation (9.81), we get

$$I = \frac{V\left[g_m + \dfrac{1}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]} \tag{9.88}$$

Using this relation in Equation (9.87), we get

$$V_o = \frac{V\left[g_m + \dfrac{1}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]} . R_D$$

Substuting for V from Equation (9.86), we get

$$V_o = \frac{V_i\left[g_m R_D + \dfrac{R_D}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]}$$

Now $$A_V = \frac{V_o}{V_i} = \frac{\left[g_m R_D + \dfrac{R_D}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]} \tag{9.89}$$

The positive sign for A_V reveals that V_o and V_i are inphase. Observe that, common gate configuration is a noninverting amplifier.

Expression for Output Impedance [Z_o]

To find Z_o, first we set V_i to zero. This results in the following changes in the circuit of Fig. 9.31.

- R_S gets short circuited
- $V_{gs} = -V_i = 0$

$\Rightarrow$ $g_m V_{gs} = 0$

Current source gets open circuited.
The resulting circuit is shown in Fig. 9.33.

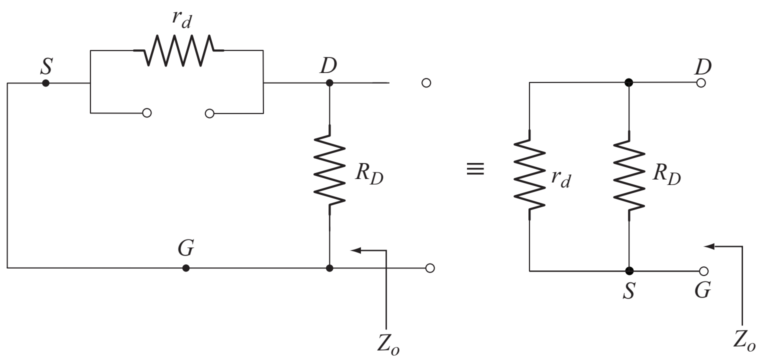


Fig. 9.33 Circuit to Find Z_o

From Fig. 9.33 we find that

$$Z_o = r_d \parallel R_D \tag{9.90}$$

Expressions for Z_i, A_V and Z_o neglecting the effect of r_d

For $r_d \geq 10\ R_D$

Consider Equation (9.81).

$$\text{Since} \quad \frac{R_D}{r_d} \leq 0.1$$

$$\therefore \quad 1 + \frac{R_D}{r_d} \approx 1$$

$$\text{Also} \quad \frac{1}{r_d} \ll g_m \quad [\text{Since } r_d \text{ is large}]$$

$$\text{As a result,} \quad g_m + \frac{1}{r_d} \approx g_m$$

Using these approximations in Equation (9.81), we get

$$Z_i' \approx \frac{1}{g_m} \tag{9.91}$$

$$\text{Now} \quad Z_i = R_S \parallel Z_i'$$

$$Z_i = R_S \parallel \frac{1}{g_m} \tag{9.92}$$

Now Consider Equation (9.89)

$$g_m R_D + \frac{R_D}{r_d} \approx g_m R_D$$

$$\text{and} \quad 1 + \frac{R_D}{r_d} \approx 1$$

$$\therefore \qquad A_V \approx g_m R_D \tag{9.93}$$

From Equation (9.90)

$$Z_o = r_d \| R_D \approx R_D \tag{9.94}$$

Important characteristics of common gate configuration

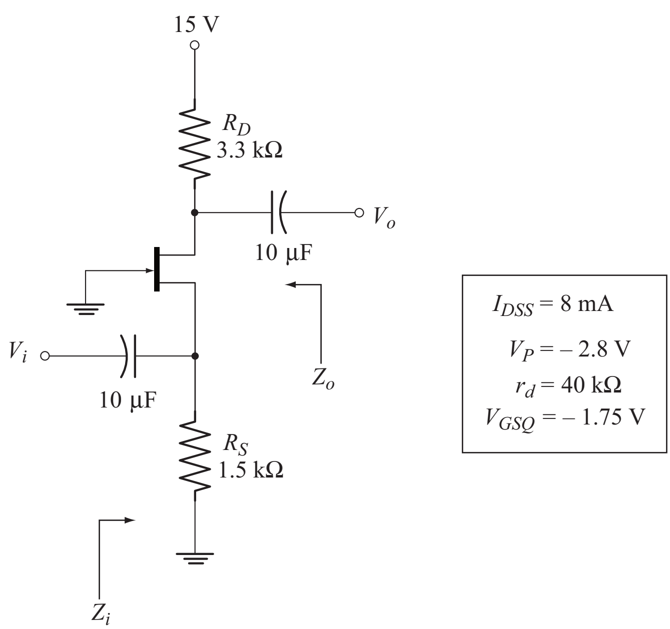
Based on the results derived, we can attribute the following characteristics to the common gate configuration.

- Relatively low input impedance $\left[Z_i \approx R_S \| \dfrac{1}{g_m}\right]$
- Medium voltage gain $[A_V \approx g_m R_D]$
- Medium high output impedance $[Z_o \approx R_D]$
- Zero phase difference between input and output voltages [non inverting amplifier]

Example 9.13

For the JFET common-gate configuration shown below:

(a) Calculate Z_i and Z_o
(b) Find A_V
(c) Calculate V_o if V_i = 1 mV(p – p)
(d) Calculate Z_i, Z_o, A_V and V_o, neglecting the effect of r_d.



Solution

$$g_m = g_{mo}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{(2)(8\text{ mA})}{2.8\text{V}} = 5.71\text{ mS}$$

$$g_m = 5.71\text{ mS}\left[1 - \frac{-1.75\text{V}}{-2.8\text{V}}\right] = 2.14\text{ mS.}$$

(a)
$$Z_i = R_S \,||\, \left[\frac{r_d + R_D}{1 + g_m r_d}\right]$$

$$= 1.5\text{ k}\Omega \,||\, \left[\frac{40\text{k}\Omega + 3.3\text{k}\Omega}{1 + (2.14\text{mS})(40\text{k}\Omega)}\right]$$

$$= 1.5\text{ k}\Omega \,||\, 0\,.5\text{ k}\Omega = 375\Omega$$

$$Z_o = r_d \,||\, R_D = 40\text{ k}\Omega \,||\, 3.3\text{ k}\Omega = 3.04\text{ k}\Omega$$

(b)
$$A_V = \frac{\left[g_m R_D + \dfrac{R_D}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]}$$

$$= \frac{\left[(2.14\text{mS})(3.3\text{k}\Omega) + \dfrac{3.3\text{k}\Omega}{40\text{k}\Omega}\right]}{\left[1 + \dfrac{3.3\text{k}\Omega}{40\text{k}\Omega}\right]} = \frac{7.1445}{1.0825} = 6.6$$

(c)
$$V_o = A_V V_i = [6.6]\,[1\text{mV}] = 6.6\text{ mV(p – p)}$$

(d)
$$r_d = 40\text{ k}\Omega \quad \text{and} \quad 10\,R_D = 3.3\text{ k}\Omega$$

Since $r_d > 10\,R_D$, the effect of r_d can be neglected.

$$Z_i = R_S \,||\, \frac{1}{g_m} = 1.5\text{ k}\Omega \,||\, \frac{1}{2.14\text{mS}} = 356.12\Omega$$

$$Z_o = R_D = 3.3\text{ k}\Omega$$

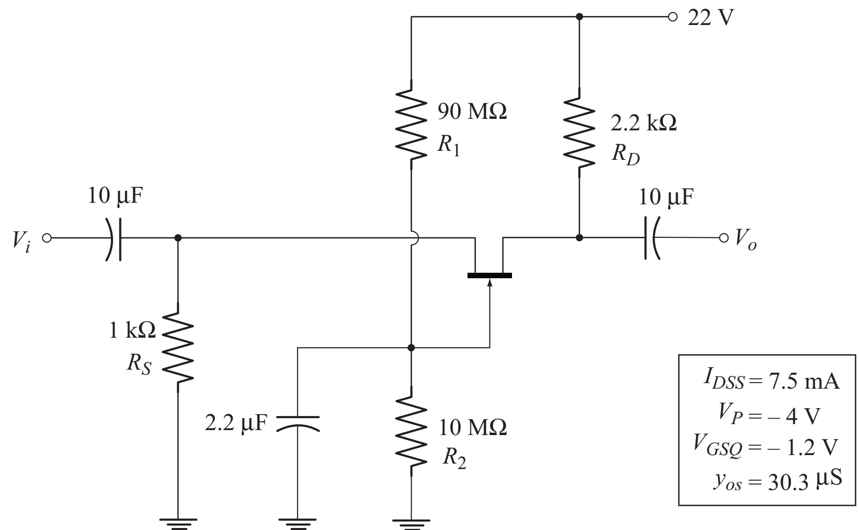
$$A_V = g_m R_D = [2.14\text{ mS}]\,[3.3\text{ k}\Omega] = 7.06$$

$$V_o = A_V V_i = [7.06]\,[1\text{ mV}] = 7.06\text{ mV(p – p)}$$

Example 9.14

For the circuit shown below:

(a) Calculate Z_i and Z_o.
(b) Find A_V.
(c) Calculate V_o, if $V_i = 5$ mV (rms).
(d) Calculate Z_i, Z_o, A_V and V_o, neglecting the effect of r_d.



Solution

The given circuit is JFET common-gate configuration using voltage divider bias. The 22 μF capactior is used to create ground at the gate terminal for ac operation. As a result, the voltage divider resistors R_1 and R_2 will get shorted to ground.

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{(2)(7.5\,\text{mA})}{4\,\text{V}} = 3.75\text{ mS}$$

$$g_m = 3.75\text{ mS}\left[1 - \frac{-1.2\,\text{V}}{-4\,\text{V}}\right] = 2.625\text{ mS}$$

$$r_d = \frac{1}{y_{o_s}} = \frac{1}{30.3\,\mu\text{S}} = 33\text{ k}\Omega$$

(a)

$$Z_i = R_S \,||\, \left[\frac{r_d + R_D}{1 + g_m r_d}\right]$$

$$= 1\text{ k}\Omega \,||\, \left[\frac{33\,\text{k}\Omega + 2.2\,\text{k}\Omega}{1 + (2.625\,\text{mS})(33\,\text{k}\Omega)}\right] = 286.58\ \Omega$$

$$Z_o = r_d \,||\, R_D = 33\text{ k}\Omega \,||\, 2.2\text{ k}\Omega = 2.06\text{ k}\Omega$$

(b)

$$A_V = \frac{\left[g_m R_D + \dfrac{R_D}{r_d}\right]}{\left[1 + \dfrac{R_D}{r_d}\right]}$$

$$= \frac{\left[(2.625\,\text{mS})\ (2.2\,\text{k}\Omega)\ +\ \dfrac{2.2\,\text{k}\Omega}{33\,\text{k}\Omega}\right]}{\left[1+\dfrac{2.2\,\text{k}\Omega}{33\,\text{k}\Omega}\right]} = 5.47$$

(c) $V_o = A_V V_i = [5.47]\ [5\text{ mV}] = 27.35\text{ mV (rms)}.$

(d) $r_d = 33\text{ k}\Omega$ and $10\,R_D = (10)\,(2.2\text{ k}\Omega) = 22\text{ k}\Omega$

Since $r_d > 10\,R_D$, the effect of r_d can be neglected

$$Z_i = R_S \,||\, \frac{1}{g_m} = 1\text{k}\Omega \,||\, \frac{1}{2.625\,\text{mS}} = 275.86\Omega$$

$$Z_o = R_D = 2.2\text{ k}\Omega$$

$$A_V = g_m R_D = (2.625\text{ mS})\,(2.2\text{ k}\Omega) = 5.775$$

$$V_o = A_V V_i = [5.775]\ [5\text{ mV}] = 28.875\text{ mV (rms)}$$

9.16 DEPLETION-TYPE MOSFETS [D-MOSFET]

MOSFET stands for metal oxide semiconductor field effect transistor. The name metal oxide is due to the fact that, the gate is made up of metal and a thin insulating silicon dioxide (SiO_2) layer separates the gate from the channel. Since the gate is electrically insulated from the channel, MOSFET has very high input impedance which is of the same order as that of JFET. As a result the gate current I_G is essentially zero for the dc-biased configurations.

The drain current and transconductance equations derived for JFET can be readily applied to D-MOSFET. These equations are reproduced below for reference.

$$I_D = I_{DSS}\left[1-\frac{V_{GS_Q}}{V_p}\right]^2 \tag{9.95}$$

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right] \tag{9.96}$$

where $$g_{m_o} = \frac{2\,I_{DSS}}{|V_p|} \tag{9.97}$$

In JFET, V_{GS_Q} can only be negative for n channel devices and positive for p channel devices. Hence g_m is always less than g_{m_o}. But for D-MOSFET, V_{GS_Q} can be of either polarily.

For instance, if V_{GS_Q} is negative for an n channel device, the device operates in the depletion mode. In depletion mode, $g_m < g_{m_o}$ and $I_D < I_{DSS}$ as seen from equations (9.95) and (9.96).

On the other hand, if V_{GS_Q} is positive, the device operates in the enhancement mode. In enhancement mode $g_m > g_{m_o}$ and $I_D > I_{DSS}$.

Small Signal Model of D-MOSFET

The small signal ac model of D-MOSFET is exactly the same as that of JFET. The range of r_d is same as that of JFET. The only difference is in the value of g_m. Figure 9.34 shows the circuit symbol and small signal ac model of D-MOSFET.

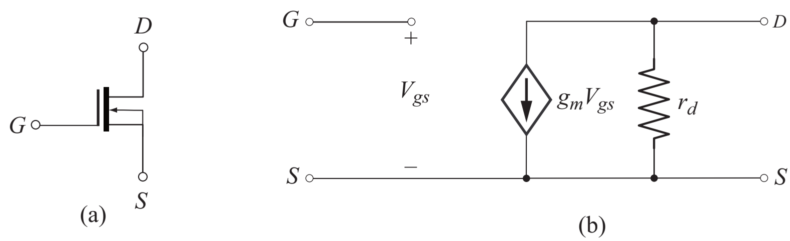


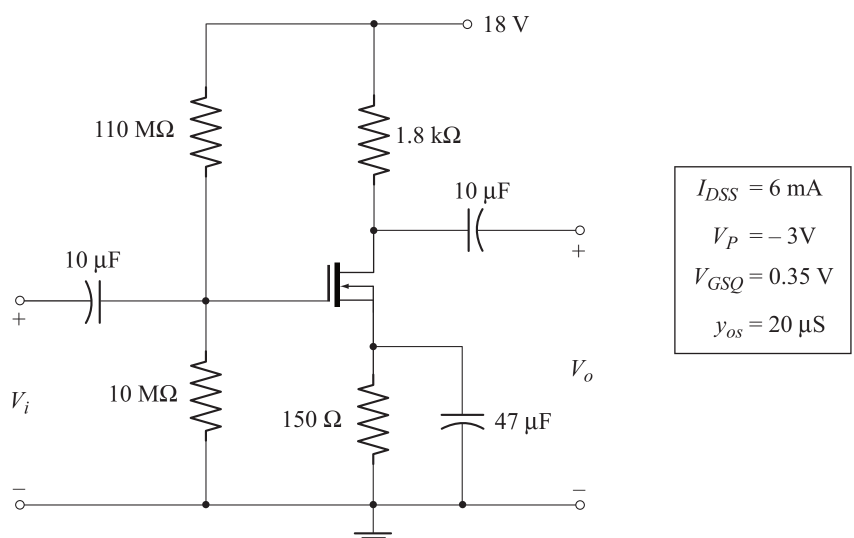
Fig. 9.34 (a) Circuit symbol of n channel D-MOSFET
(b) Small-signal ac model of D-MOSFET

It should be noted that the small signal ac model is same for both *n*-channel and p-channel D-MOSFETS. Also all the results derived for JFET configurations can be readily applied to the corresponding D-MOSFET configurations.

Example 9.15

For the D-MOSFET amplifier shown

(a) Calculate g_m and compare with g_{mo}
(b) Calculate r_d
(c) Draw the ac equivalent circuit
(d) Find Z_i and Z_o
(e) Calculate A_V
(f) Find V_o if V_i = 2 mV(p – p)



Solution

The given circuit is D-MOSFET common-source configuration using voltage divider bias. The results derived in section 9.12 for JFET configuration can be directly used for this circuit.

(a)
$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

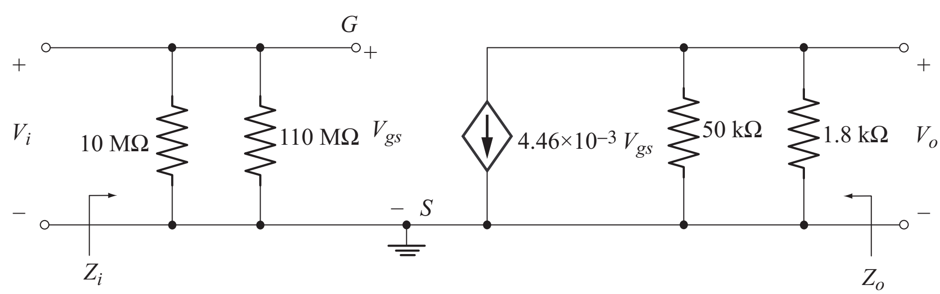
$$g_{m_o} = \frac{2I_{DSS}}{|V_p|} = \frac{(2)(6\,\text{mA})}{3\,\text{V}} = 4\text{ mS}$$

$$g_m = 4\text{ mS}\left[1 - \frac{0.35\,\text{V}}{-3\,\text{V}}\right] = 4.46\text{ mS}$$

Note that, $g_m > g_{m_o}$ since V_{GS_Q} is positive.

(b)
$$r_d = \frac{1}{y_{os}} = \frac{1}{20\,\mu\text{S}} = 50\text{ k}\Omega$$

(c) The ac equivalent circuit is shown below.



(d) From Equation (9.49),

$$Z_i = R_1 \,\|\, R_2$$
$$= 110\text{ M}\Omega \,\|\, 10\text{ M}\Omega = 9.16\text{ M}\Omega$$

From Equation (9.50),

$$Z_o = r_d \,\|\, R_D = 50\text{ k}\Omega \,\|\, 1.8\text{ k}\Omega = 1.73\text{ k}\Omega$$

(e) From Equation (9.51),

$$A_V = -g_m\,[r_d \,\|\, R_D]$$
$$= -[4.46\text{ mS}]\,[50\text{ k}\Omega \,\|\, 1.8\text{ k}\Omega] = -7.71$$

(f)
$$V_o = A_V V_i = [-7.71]\,[2\text{ mV}] = -15.42\text{ mV}_{(p-p)}$$

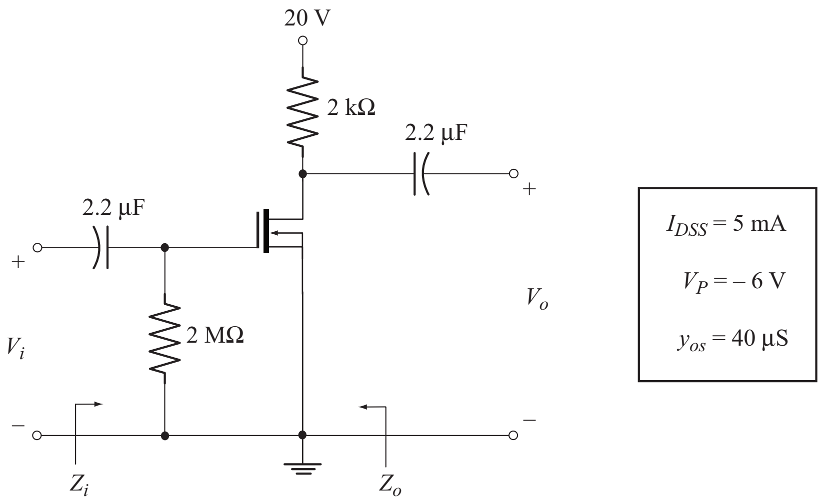
Example 9.16

For the D-MOSFET amplifier shown below

(a) Calculate Z_i and Z_o
(b) Find A_V
(c) Calculate V_o if $V_i = 50$ mV (rms)
(d) Find Z_i, Z_o, A_V and V_o neglecting the effect of r_d.

Solution

The given circuit is a D-MOSFET common source configuration with $V_{GS_Q} = 0$ V. For the solution, please refer Example 9.5.



Example 9.17

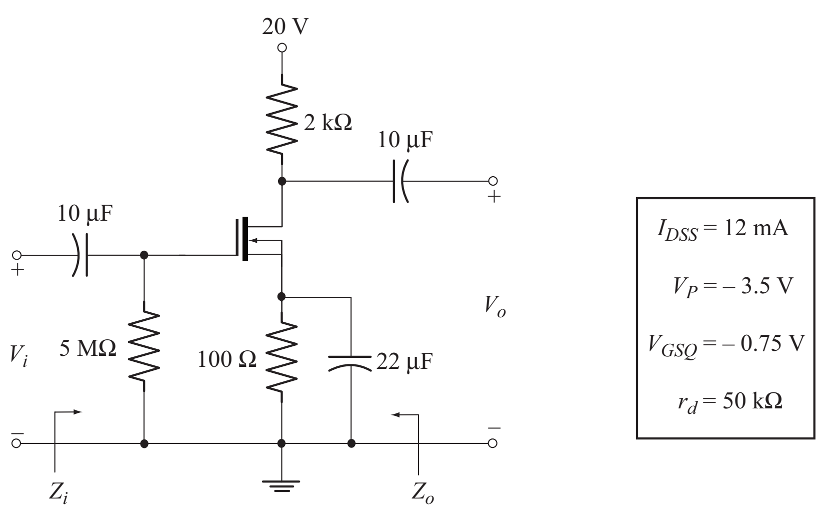
For the circuit shown below

(a) Calculate Z_i and Z_o

(b) Calculate A_V

(c) Find V_o if V_i = 1 mV (rms)

(d) Determine Z_i, Z_o, A_V and V_o neglecting the effect of r_d.



Solution

The given circuit is D-MOSFET common-source configuration using self bias. The results derived in section 9.10 for JFET configuration can be readily applied to this circuit.

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_{m_o} = \frac{2 I_{DSS}}{|V_p|}$$

$$= \frac{2\,(12\,\text{mA})}{3.5\,\text{V}} = 6.85\text{ mS}$$

$$g_m = 6.85\text{ mS}\left[1 - \frac{-0.75\,\text{V}}{-3.5\,\text{V}}\right] = 5.38\text{ mS}$$

(a) $$Z_i = R_G = 5\text{ M}\Omega \qquad \text{[From Equation (9.26)]}$$

$$Z_o = r_d \parallel R_D \qquad \text{[From Equation (9.27)]}$$

$$= 50\text{ k}\Omega \parallel 2\text{ k}\Omega = 1.92\text{ k}\Omega$$

(b) $$A_V = -g_m\,[r_d \parallel R_D] \qquad \text{[From Equation (9.28)]}$$

$$= -[5.\,38\text{ mS}]\,[1.92\text{ k}\Omega]$$

$$= -10.33.$$

(c) $$V_o = A_V V_i$$

$$= [-10.33]\,[1\text{ mV}]$$

$$= -10.33\text{ mV(rms)}$$

(d) $$r_d = 50\text{ k}\Omega \quad \text{and} \quad 10\,R_D = 20\text{ k}\Omega$$

Since $r_d > 10\,R_D$, the effect of r_d can be neglected.

$$Z_i = R_G = 5\text{ M}\Omega$$

$$Z_o = R_D = 2\text{ k}\Omega$$

$$A_V = -g_m R_D$$

$$= -[5.38\text{ mS}]\,[2\text{ k}\Omega] = -10.76$$

$$V_o = [-10.76]\,[1\text{ mV}]$$

$$= -10.76\text{ mV (rms)}$$

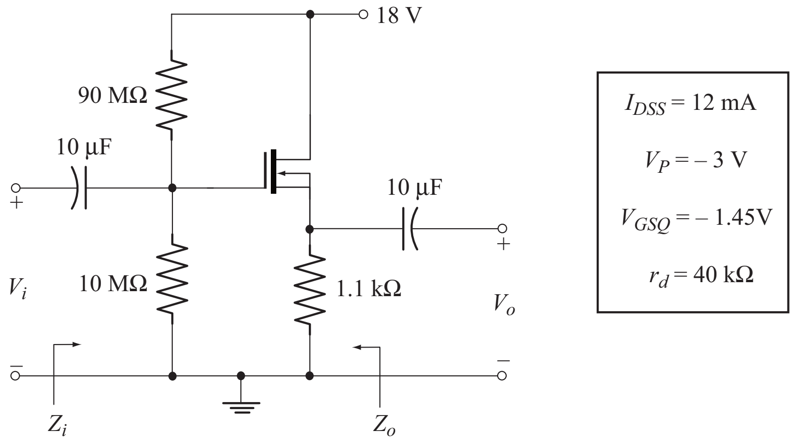
Example 9.18

For the amplifier shown below:

(a) Calculate Z_i and Z_o

(b) Calculate A_V

(c) Determine V_o if $V_i = 2$ mV (rms)



Solution

The given circuit is D-MOSFET source-follower configuration without R_D. The results derived in section 9.14 for JFET source-follower can be readily applied to this circuit.

$$g_m = g_{m_o}\left[1 - \frac{V_{GS_Q}}{V_p}\right]$$

$$g_{m_o} = \frac{2I_{DSS}}{|V_p|}$$

$$= \frac{2\,(12\,\text{mA})}{3\,\text{V}} = 8 \text{ mS}$$

$$g_m = 8 \text{ mS}\left[1 - \frac{-1.45\,\text{V}}{-3\,\text{V}}\right] = 4.13 \text{ mS}$$

(a) $$Z_i = 90 \text{ M}\Omega \,||\, 10 \text{ M}\Omega = 9 \text{ M}\Omega$$

$$Z_o = \frac{1}{g_m} \,||\, r_d \,||\, R_S \qquad \text{[From Equation (9.72)]}$$

$$= \frac{1}{4.13\,\text{mS}} \,||\, 40 \text{ k}\Omega \,||\, 1.1 \text{ k}\Omega = 197.46\Omega$$

(b) $$A_V = \frac{g_m\,[r_d \,||\, R_S]}{1 + g_m\,[r_d \,||\, R_S]} \qquad \text{[From Equation (9.65)]}$$

$$= \frac{[4.13\,\text{mS}][40\,\text{k}\Omega \,||\, 1.1\,\text{k}\Omega]}{1 + [4.13\,\text{mS}][40\,\text{k}\Omega \,||\, 1.1\,\text{k}\Omega]} = 0.815$$

(c) $$V_o = A_V\,V_i$$

$$= [0.815]\,[2 \text{ mV}] = 1.63 \text{ mV (rms)}$$

◆ 9.17 ENHANCEMENT-TYPE MOSFETS [E-MOSFET]

Figure 9.35 shows the circuit symbols of n-channel (n MOS) and p-channel (p MOS) E-MOSFETS. The E-MOSFET is fabricated to operate only in enhancement mode. Hence V_{GS_Q} can only be positive for n-channel device and negative for p-channel device.

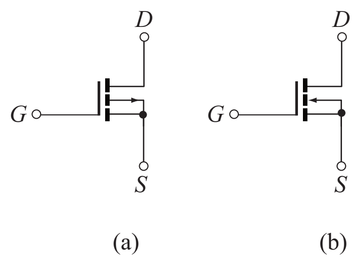


Fig. 9.35 (a) Circuit symbol of p MOS device
(b) Circuit symbol n MOS device

The drain current for E-MOSFET is given by the relation

$$I_D = k\,[V_{GS} - V_{GS\,(\mathrm{Th})}]^2 \tag{9.98}$$

$V_{GS(\mathrm{Th})}$ is the threshold voltage. It is important to note that, the drain current is zero for $V_{GS} < V_{GS(\mathrm{Th})}$ and it significantly increases for $V_{GS} \geq V_{GS(\mathrm{Th})}$. k is a constant which is specified on the data sheet for typical operating point.

◆ 9.18 EXPRESSION FOR TRANSCONDUCTANCE g_m

The transconductance is defined by

$$g_m = \left.\frac{dI_D}{d\,V_{GS}}\right|_{Q\,\mathrm{Point}} \tag{9.99}$$

Differentiating Equation (9.98) with respect to V_{GS}, we get

$$\frac{dI_D}{dV_{GS}} = 2\,k\,[V_{GS} - V_{GS(\mathrm{Th})}]$$

But $$g_m = \left.\frac{dI_D}{d\,V_{GS}}\right|_{Q\ \mathrm{Point}}$$

$$\therefore \quad g_m = 2\,k\,[V_{GSQ} - V_{GS(\mathrm{Th})}] \tag{9.100}$$

◆ 9.19 SMALL SIGNAL AC MODEL OF E-MOSFET

The small signal ac model of E-MOSFET is shown in Fig. 9.36. Observe that, the format of the model is exactly identical to that of JFET.

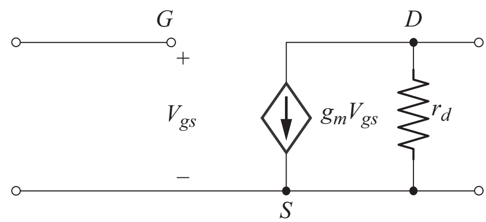


Fig. 9.36 Small Signal ac model of E-MOSFET

◆ 9.20 E-MOSFET DRAIN-FEEDBACK CONFIGURATION

Figure 9.37 shows the circuit of E-MOSFET common-source amplifer employing drain feedback. The resistor R_F provides ac negative feedback.

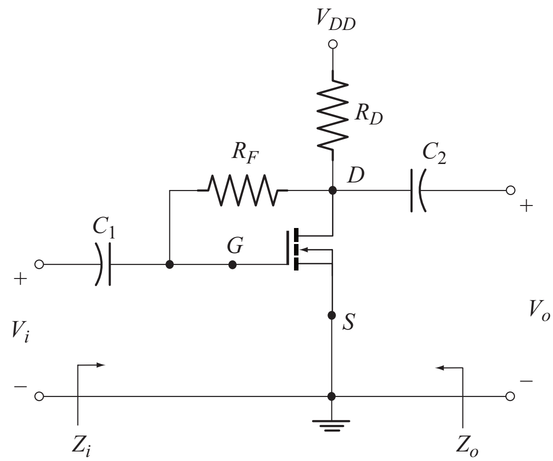


Fig 9.37 E – MOSFET drain-feedback configuration

The ac equivalent circuit is drawn in Fig. 9.38 by reducing V_{DD} to zero and replacing C_1 and C_2 with their short circuit equivalents.

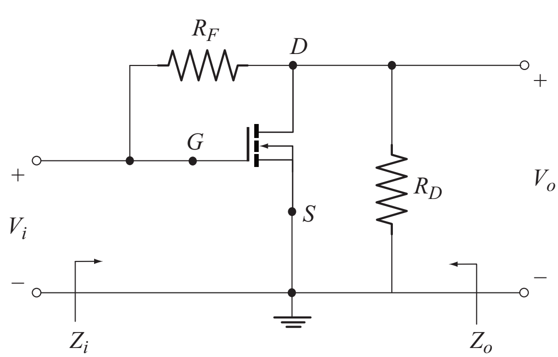


Fig 9.38 AC equivalent circuit of E-MOSFET drain-feedback configuration

The ac equivalent circuit is redrawn in Fig. 9.39 by replacing E–MOSFET with its small–signal ac model.

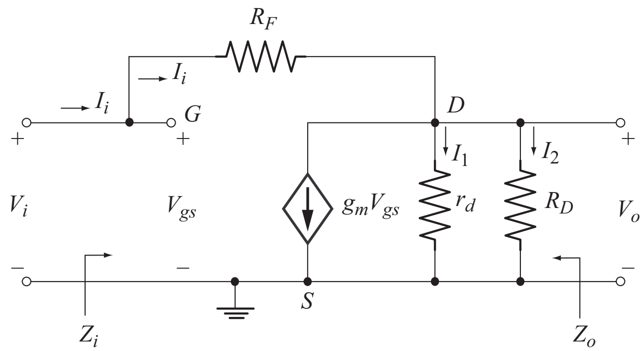


Fig 9.39 Ac equivalent circuit using small signal ac model of E-MOSFET

Due to open circuit between the gate and source terminals, the current into the gate terminal is zero. Hence I_i flows into R_F.

Expression for input impedance [Z_i]

Due to the connection between input and output nodes, the calculation Z_i is not straight forward. The input impedance is given by

$$Z_i = \frac{V_i}{I_i} \tag{9.101}$$

Applying KCL at the node D, we have

$$I_i = g_m V_{gs} + I_1 + I_2 \tag{9.102}$$

From the input circuit

$$V_{gs} = V_i \tag{9.103}$$

From the output circuit,

$$I_1 = \frac{V_o}{r_d} \quad \text{and} \quad I_2 = \frac{V_o}{R_D}$$

Now $$I_1 + I_2 = \frac{V_o}{r_d} + \frac{V_o}{R_D}$$

$$= V_o\left[\frac{1}{r_d} + \frac{1}{R_D}\right]$$

$$= V_o\left[\frac{1}{R'_D}\right] \tag{9.104}$$

where $$\frac{1}{R'_D} = \frac{1}{r_d} + \frac{1}{R_D}$$

$$\Rightarrow \qquad R'_D = \frac{r_d R_D}{r_d + R_D} = r_d \parallel R_D \tag{9.105}$$

Using Equations (9.103) and (9.104) in Equation (9.102), we get

$$I_i = g_m V_i + \frac{V_o}{R'_D} \tag{9.106}$$

$$\text{But} \qquad I_i = \frac{V_i - V_o}{R_F} \qquad [\text{current through } R_F]$$

$$\Rightarrow \qquad V_o = V_i - I_i R_F \tag{9.107}$$

Using Equation (9.107) in Equation (9.106), we have

$$I_i = g_m V_i + \frac{V_i - I_i R_F}{R'_D}$$

$$I_i \left[1 + \frac{R_F}{R'_D} \right] = V_i \left[g_m + \frac{1}{R'_D} \right]$$

$$\text{Now} \qquad Z_i = \frac{V_i}{I_i} = \frac{\left[1 + \frac{R_F}{R'_D} \right]}{\left[g_m + \frac{1}{R'_D} \right]}$$

$$\text{or} \qquad Z_i = \frac{R_F + R'_D}{1 + g_m R'_D} \tag{9.108}$$

$$\text{or} \qquad Z_i = \frac{R_F + [r_d \parallel R_D]}{1 + g_m [r_d \parallel R_D]} \tag{9.109}$$

Approximate expression for Z_i

Usually, $R_F >> r_d \parallel R_D$

$$\therefore \qquad R_F + [r_d \parallel R_D] \approx R_F$$

Using this approximation in Equation (9.109) we get

$$Z_i \approx \frac{R_F}{1 + g_m [r_d \parallel R_D]}$$

$$\text{Further, if} \qquad r_d \geq 10\, R_D,$$

$$r_d \parallel R_D \approx R_D$$

Now we get

$$Z_i \approx \frac{R_F}{1+g_m R_D} \tag{9.110}$$

Expression for Voltage Gain [A_V]

From Equation (9.106) we have

$$I_i = g_m V_i + \frac{V_o}{R'_D} \tag{9.111}$$

Using $\quad I_i = \dfrac{V_i - V_o}{R_F}$, we get

$$\frac{V_i - V_o}{R_F} = g_m V_i + \frac{V_o}{R'_D}$$

$$V_o \left[\frac{1}{R_F} + \frac{1}{R'_D}\right] = V_i \left[\frac{1}{R_F} - g_m\right]$$

$$A_V = \frac{V_o}{V_i} = \frac{\left[\dfrac{1}{R_F} - g_m\right]}{\left[\dfrac{1}{R_F} + \dfrac{1}{R'_D}\right]} \tag{9.112}$$

$$\frac{1}{R'_D} = \frac{1}{r_d} + \frac{1}{R_D}$$

$$\therefore \quad \frac{1}{R_F} + \frac{1}{R'_D} = \frac{1}{R_F} + \frac{1}{r_d} + \frac{1}{R_D}$$

$$\Rightarrow \quad \frac{1}{\dfrac{1}{R_F} + \dfrac{1}{R'_D}} = R_F \,||\, r_d \,||\, R_D \tag{9.113}$$

Using Equation (9.113) in Equation (9.112) we get

$$A_V = \left[\frac{1}{R_F} - g_m\right][R_F \,||\, r_d \,||\, R_D] \tag{9.114}$$

Usually $\dfrac{1}{R_F}$ is smaller than g_m. Hence A_V is negative. The negative sign reveals that, V_o and V_i are out of phase by 180°.

Approximate Expression for A_V

Typically $\dfrac{1}{R_F} \ll g_m$. As a result

$$\frac{1}{R_F} - g_m \approx -g_m$$

Also $$R_F \gg r_d \,||\, R_D$$

Hence $$R_F \,\|\, r_d \,\|\, R_D \approx r_d \,\|\, R_D$$

Using these approximations in Equation (9.114) we get

$$A_V \approx -g_m\,[r_d \,||\, R_D]$$

If $r_d \geq 10\,R_D$, then $r_d \,||\, R_D \approx R_D$

Now we get $$A_V \approx -g_m\,R_D \tag{9.115}$$

Expression for Output Impedance [Z_o]

To find the output impedance, we reduce V_i to zero in the circuit of Fig 9.39. With $V_i = 0$

- Gate get shorted to source
- $g_m V_{gs} = 0$. Therefore the current $g_m V_{gs}$ represents an open circuit.

The resulting circuit is shown in Fig 9.40.

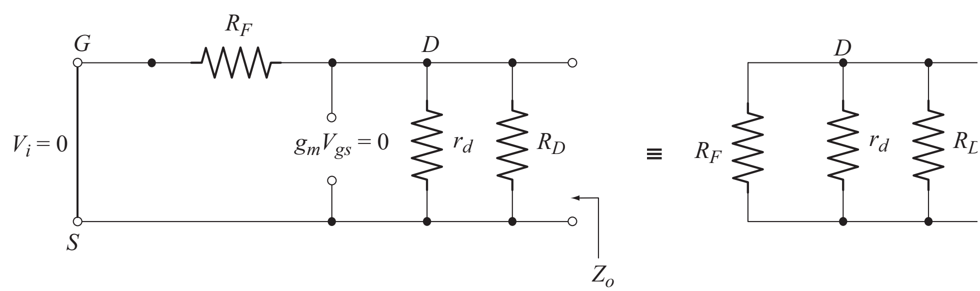


Fig 9.40 Circuit to find Z_o

From Fig. 9.40 we find that

$$Z_o = R_F \,||\, r_d \,||\, R_D \tag{9.116}$$

Approximate Expression for Z_o

Using the conditions

$R_F \gg r_d \,||\, R_D$ and $r_d \geq 10\,R_D$ in Equation (9.116) we get

$$Z_o \approx R_D \tag{9.117}$$

Example 9.19

A MOSFET has $V_{GS\,(\mathrm{Th})} = 3.5$ V and it is biased at $V_{GS_Q} = 7$ V. Assuming $k = 0.5 \times 10^{-3}$ A/V^2 calculate the value of g_m.

Solution

$$g_m = 2\,k\,[V_{GS_Q} - V_{GS\,(\mathrm{Th})}]$$
$$= (2)\,(0.5 \times 10^{-3}\ \mathrm{A/V^2})\,[7\mathrm{V} - 3.5\mathrm{V}] = 3.5\ \mathrm{mS}.$$